SE499

Software Engineering Design Development project

Department of Computer Science

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SRS

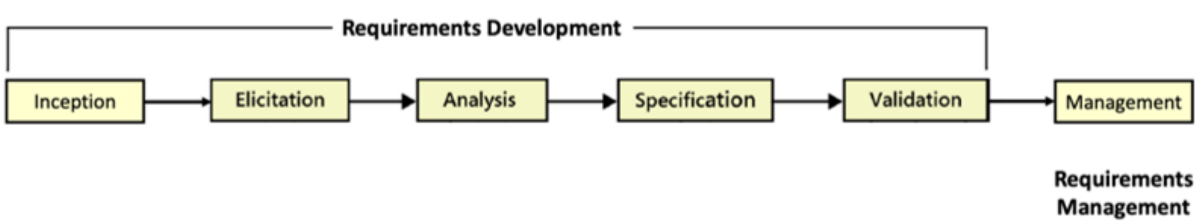
SOFTWARE REQUIREMENT SPECIFICATION PROCESS

Home Key

Introduction:

The purpose of this document is to provide a comprehensive overview of the software engineering process. Given that our website concept is not entirely original, we undertook extensive research by examining other websites with similar functionalities. This involved reading reviews, assessing ratings, studying user feedback, and considering various opinions. While maintaining core features found in existing platforms, we strategically incorporated additional features that we identified as crucial for enhancing the user experience. We conducted a focus group involving potential renters and property bookers with prior experience in using booking platforms. Then, in our second iteration, we conducted an online workshop targeting individuals who frequently engage with similar applications. These insights were invaluable in refining our platform. Furthermore, we thoroughly analyzed and prioritized the project requirements. This document serves as a guide for the project team members responsible for executing and validating the requirements engineering process.

Requirements Engineering Process:



First Iteration:

1. Elicitation

In the initial elicitation phase of the requirement engineering process's first iteration, we organized a focus group session with individuals who were enthusiastic users of property booking applications. The objective was to gain insights into their needs and collect information about essential requirements. Our primary focus was to comprehend the property booking process and identify the specific features they considered crucial. During these sessions, participants openly shared their thoughts, opinions, and ideas, providing a comprehensive perspective on the subject matter.

During the gathering, we obtained their consent to record the meeting for future reference, and provided a brief overview of the website's concept, even though we knew that the majority of the participants were already familiar with these applications. We then presented a Google Form containing a series of questions, outlining our plan and approach for addressing these inquiries. Our strategy was to address one question at a time, engaging in discussions afterward to ensure that everyone had an opportunity to express their views, from which we took detailed notes for reference. For each question, participants completed the corresponding section in the Google Form and subsequently shared their responses verbally. They elaborated on their choices, offered insights into their perspectives, and sometimes even highlighted differing viewpoints. This approach facilitated productive discussions and provided us with profound insights. We maintained a consistent sequence of addressing, discussing, and transitioning to the subsequent question for all inquiries.

We utilized Google Form to document all responses, creating a valuable record that enabled us to gather and analyze the responses. During this process, we allocated our responsibilities as follows: Tala and Leena were responsible for questioning and engaging with the participants, Batoul and Shaikah were tasked with taking notes, and Haifa was in charge of documenting the proceedings and verifying that all focus group needs were addressed. Following the focus group, we assembled a comprehensive list of website functionalities. Survey questions and responses can be found in the appendix.

2. Analysis

After we gathered information from the first step, we started looking closely at it. The purpose of this is to sort the information into different categories and fix any problems like things that are not clear, missing parts, things that don't match, or things that were left out. In this phase, we carefully checked what the end users told us. We wanted to make sure that all the information was complete and could actually work. We also looked for any mistakes or things that didn't fit in the information we got. We used one method to analyze our requirements, which was by using activity diagrams.

3. Specification

During this phase, we began documenting all the user needs in a written document. We started with a broad view and divided the requirements into two categories: functional and nonfunctional. This helped specify the important features the system should have. As a team, we aimed to avoid creating numerous separate statements and refrained from using negative requirements. Additionally, we made sure to highlight any crucial points that needed further clarification, knowing that we had a second iteration and another data collection technique to be employed. We recorded these requirements in the SRS document, which served as the foundation for all subsequent development tasks.

4. Validation

In the validation stage, our focus was on verifying the SRS document to guarantee the accuracy, completeness, and absence of errors in all the requirements. As a team, we thoroughly examined the document and confirmed that the requirements aligned with their needs. Additionally, the validation technique used was reviews where reconnected with the end users to ensure that the requirements were comprehensive and left no gaps.

Second Iteration:

1. Elicitation

In the second iteration of the requirements engineering process, we encountered a shortage of requirements to document. As a result, we recognized the need for an additional elicitation technique to ensure comprehensive coverage. Consequently, we chose to initiate a second iteration of the requirements engineering process. For the elicitation phase in this second iteration, we conducted an online workshop through google meet, aimed at gathering new requirements. Our approach involved inviting the original five participants from the focus group and adding an additional five individuals. In total, we received feedback from 10 participants, all of whom met the condition of having prior experience with the applications. This condition was crucial to gather more accurate and precise insights from actual users. Our method involved using a Google form with concise 2–3-word responses, prioritization, and verbal discussions. Overall, this approach proved effective, yielding valuable information. Survey questions and responses can be found in the appendix.

2. Analysis

Following the second round of gathering requirements, we once again examined our list of needs. The aim of requirements analysis remains consistent: it involves studying the information collected during the elicitation phase, organizing it into different groups, and resolving any issues related to ambiguity, vagueness, omissions, or contradictions in the identified needs. We used one method to analyze our requirements, which was by using activity diagrams as well.

3. Specification

During this stage, we integrated the newly gathered features from the survey into the existing ones and made corresponding updates to the SRS document. Our objective was not only to present the requirements cohesively but also to maintain a positive and user-oriented tone throughout the document. We were diligent in guaranteeing that all requirements were in harmony and clearly defined to provide a solid foundation for the development team. This meticulous approach aimed to enhance the overall clarity and coherence of the document, ensuring that it serves as a reliable guide for the project's successful execution and validation.

4. Validation

The validation method we utilized entailed conducting reviews where we reengaged with the end users. Our primary objective was to meticulously examine the requirements, leaving no room for gaps or ambiguities. Through this iterative process, we maintained an ongoing dialogue with the end users, consistently incorporating their valuable perspectives into the requirements. This iterative approach resulted in a more comprehensive and user-centric final product, reinforcing our commitment to meeting the users' needs and expectations.

Requirements Management

We have implemented a change control process to oversee and align changes with project objectives and stakeholder requirements. Our primary objective within this procedure is to minimize the impact of changes on the project's timeline, work and effort all while upholding the quality of the project's deliverables.

i. Step 1: Initiating Requests:
Stakeholders initiate requests for project requirements.

ii. Step 2: Request Assessment:
The submitted request undergoes a thorough analysis to ascertain its validity, assess potential impacts on other requirements, and provide estimations for various aspects, including cost, before proceeding with the implementation of the proposed change. This meticulous evaluation helps ensure that the decision to implement the change is well-informed and takes into account all relevant factors.

iii. Step 3: Change Review:
Following further analysis and validation, the change request is presented to our team for a comprehensive review. It is during this stage that we collectively decide on the request's status, taking into account its alignment with project goals, potential impacts on timelines, and resource availability. This collaborative approach allows us to make informed decisions that prioritize the overall success of the project while maintaining transparency and inclusivity in the decision-making process.

iv. Step 4: Change Implementation & Stakeholder Communication:
In case of approval, the change is executed, and necessary updates are incorporated into the Software Requirements Specification (SRS). We promptly communicate with stakeholders to ensure their confidence in the successful implementation of the change. In the event that approval is not granted, a formal communication is issued to inform about the decision.

Requirements Prioritization

Finally, we established priority among the requirements through the utilization of a survey. This survey closely mirrored the one used during the workshop, where participants were tasked with ranking and prioritizing the specific features and contributions that the system would provide. This approach allowed us to consolidate and align our understanding of the most crucial aspects of the project according to the perspectives of those who would interact with the system.

In addition to this, we considered the survey results as a valuable input for refining our project scope and ensuring that the development efforts would be focused on delivering the most impactful features to our users. This user-centric approach, where user feedback and priorities guide the project's direction, underlines our commitment to creating a system that truly meets the needs and expectations of its users. By incorporating user perspectives and prioritizing their preferences, we aim to enhance user satisfaction, optimize resource allocation, and ultimately deliver a system that not only meets but exceeds the expectations of our stakeholders. This iterative and feedback-driven approach is integral to our commitment to delivering a successful and user-focused project.

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SRS

SOFTWARE REQUIREMENT SPECIFICATION

Home Key

1. Introduction

1.1 Purpose

The purpose of our website is to provide a platform for users to easily book, rent, and buy various types of properties, such as villas, apartments, chalets, resorts, and more, in Saudi Arabia. We aim to address the limited availability of such services in our country and serve our local community. In short, our website aims to be user-friendly, reliable, and complete. We want to simplify the process of finding and renting properties in Riyadh and other places in Saudi Arabia. Finally, We plan to introduce additional features, including various contributions.

1.2 Document Conventions

The formatting used in this document employs bold formatting to differentiate and categorize various sections and subsections; references are denoted using italicized text. Furthermore, a standardized font is employed throughout the document to maintain consistency.

1.3 Intended Audience and Reading Suggestions

This document is intended for individuals engaged in website application development, design, testing, and management, as well as for the end-users of the application. The following sections offer an in-depth exploration of the product's capabilities and attributes, emphasizing the services it delivers. The document is optimized for reading in its original layout, and each heading and subheading offers a comprehensive description of every facet of the application.

1.4 Product Scope

The proposed platform aims to create a comprehensive online tool for people looking for properties in Riyadh. This platform aims to make it easy to find properties by offering user-friendly features like advanced search, user profiles, property details, interactive maps, reviews and ratings, a mobile-friendly design, customer support, secure messaging, interior design services, and info about nearby places. These features will work together to make the *property search experience straightforward and reliable, making users happier. HomeKey will ensure that all users have all the information they need to make informed decisions in the competitive real estate market. Through continuous innovation and a commitment to user satisfaction, this platform aims to redefine the way people search for and invest in properties, setting a new standard for the real estate industry.*

1.5 References

1- *Wiegers, K., & Beatty, J. (2013). Software Requirements 3, 3rd Edition. Microsoft Press.*

2. Overall Description

2.1 Product Perspective

Our project isn't groundbreaking or completely original, there are approximately three similar projects already in existence. Nonetheless, our intention is to enhance these existing solutions by introducing valuable new features. A website that prioritizes user-friendliness and offers increased convenience and comfort to anyone seeking to book, rent or buy properties.

2.2 Product Functions

1) *Property Listings*
2) *Advanced Search*
3) *User Profiles*
4) *Property Details*
5) *Interactive Maps*
6) *Reviews and Ratings*
7) *Mobile-Friendly Design*
8) *User Support*
9) *Secure Messaging*
10) *Interior Design Services*
11. *Nearby Amenities*
12. *Inspection Team*

2.3 User Classes and Characteristics

User	*Description*	*Functionality*
Property Seekers:	*Regular users visit the website to find properties for rent, booking or buying.*	*-User Registration (User Profile Creation).* *-Property Search.* *-Property Details and Information Viewing.* *-Property Owner Personal information and Details Viewing.* *-Interior Designers Personal information and Details Viewing.* *–Secure Communication with Property Owners.* *-Secure Communication with the Interior designers (for Unfurnished Properties).* *-Rating and Feedback Submission.* *-Rating and Feedback viewing.*
Property Owners:	*Occasional users who list their properties for rent, book and buy.*	*-User Registration (User Profile Creation).* *-Property Listing.* *-Provide Property Details such as property size.* *-Share Nearby Amenities Information.* *-Secure messaging with property seekers.* *-Rating and Feedback Submission.* *-Rating and Feedback viewing.*
Interior Designers:	*Occasional users who provide interior design services.*	*-User Registration (User Profile Creation).* *-Upload their previous design work.* *-Secure communication with property seekers.* *-Rating and Feedback Submission.*

Inspection Team:	*The individuals tasked with evaluating the property and submitting their inspection report to our system.*	*-Receiving inspection requests via email from the HomeKey team.* *-Conducting property assessments.* *-Documenting inspection findings.* *-Submitting inspection reports to our team.*

2.4 Operating Environment

Homekey will function as a web-based application, accessible through web browsers such as Safari and Chrome, among others. It will have compatibility with various operating systems, including Windows, and will utilize MongoDB as its Database Management System. Homekey will be accessible across multiple platforms, including mobile phones, desktops, and tablets.

2.5 Design and Implementation Constraints

- *Users are required to have an active internet connection.*
- *The HomeKey website's front-end development will incorporate HTML, CSS, Bootstrap, and JavaScript.*
- *The HomeKey website's back-end will be developed using Node.js and MongoDB.*
- *The HomeKey website will be presented in the English language.*
- *The delivery date for the HomeKey website is scheduled for...*

2.6 User Documentation

UD-1: *The system shall incorporate an FAQ page, where responses to common questions will be provided.*

UD-2: *The system shall feature a welcoming pop-up for new users accessing the system, enabling them to understand how to use the application.*

2.7 Assumptions and Dependencies

Assumptions:

AS-1: It is assumed that users have access to the internet.
AS-2: It is assumed that users own smart devices with web browsers.
AS-3: It is assumed that users are experienced in using websites.

Dependencies:
DE-1: Users must have a functioning email address and phone number to register for a Homekey account.

3. External Interface Requirements

3.1 Hardware Interfaces

To use HomeKey, you will need a web browser and any electronic device. All the necessary hardware for internet connectivity will function as the system's hardware interface since the website is internet-based. Some examples of this hardware include modems, WAN-LAN devices, and Ethernet cross-tables.

3.2 Software Interfaces

The HomeKey website needs to connect with the database to store, retrieve, and update user data. Additionally, it should be compatible with various operating systems, including Windows and Mac OS. We employed various tools such as MongoDB for database management, Lucidchart for creating diagrams, and Visual Studio for coding, alongside other tools for system testing.

3.3Communications Interfaces

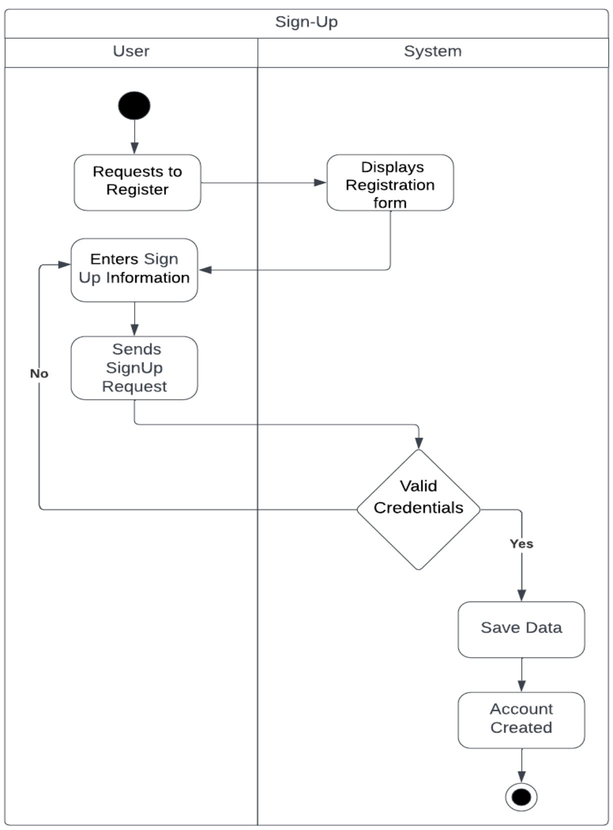
CI-1: The system shall communicate with the database that obtains all the provided information.
CI-2: The property owner and interior designers should be able to add a communication method (WhatsApp link).
CI-3: Once a booking is made by the visitor the service provider should receive a notification via Email.

4. System Features

The features of our system demonstrate its particular capabilities and functionality. These features are created to satisfy consumer demands and improve user experience. It aids users in comprehending the capabilities of the system and their potential benefits. It gives a summary of the main features that users may count on from the system. usually gives a brief explanation of each feature, along with its use and advantages. Additionally, it could explain how these functionalities can be used or accessed within the system.

4.1 System Feature 1

4.1.1 Sign-up



Description:

New users can register on the website and establish an account by supplying the necessary information. Once an account is created successfully, users will gain access to add a property and edit it, and they will be able to communicate with each other via WhatsApp or Telegram Applications.

Priority:

High priority, because without it, the website would be rendered non-functional.

4.1.2

1. **Stimulus:** User request to register

   Response Sequences: System display registration form

2. **Stimulus:** The user receives a prompt to fill in their credential

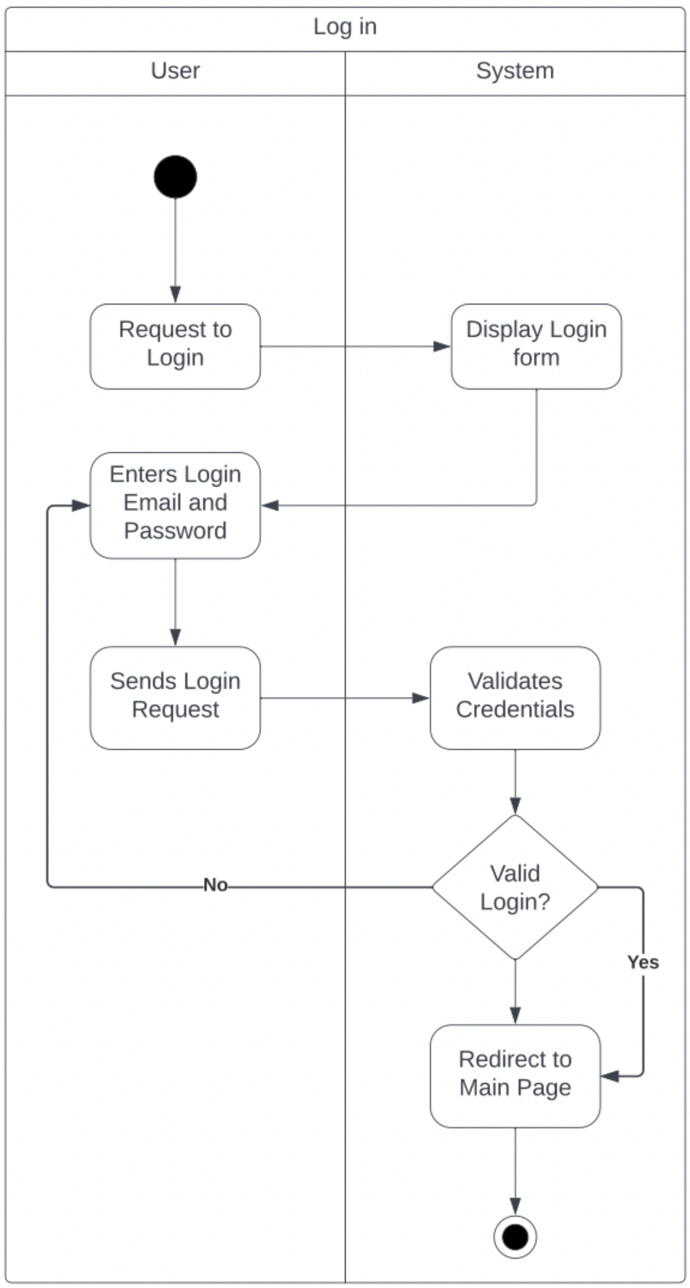
   Response Sequences: After the user provides their credential, their account is successfully created.

4.1.3

REQ-#	Functional Requirements
REQ-1	The system shall allow users to sign up.
REQ-2	The system shall allow users to register by their credentials.
REQ-3	The system shall allow users to create passwords.
REQ-4	The system shall allow users to submit their registration form
REQ-5	The system shall validate the user credentials.

4.2 System Feature 2

4.2.1 Log-in



Description:

The Users will be able to access their accounts by using their credentials, including service providers and guests.

Priority:

High priority, since this task is of critical importance because without it, the visitor cannot access their profile.

4.2.2

1. **Stimulus:** The user enters their login information to log in to the system

Response Sequences: When a user's login is valid, the system will take the user to the main page

2. **Stimulus:** The user enters their login information to log in to the system

Response Sequences: When a user’s login isn't valid the system will ask the user to re-enter her/his credentials again correctly.

4.2.3

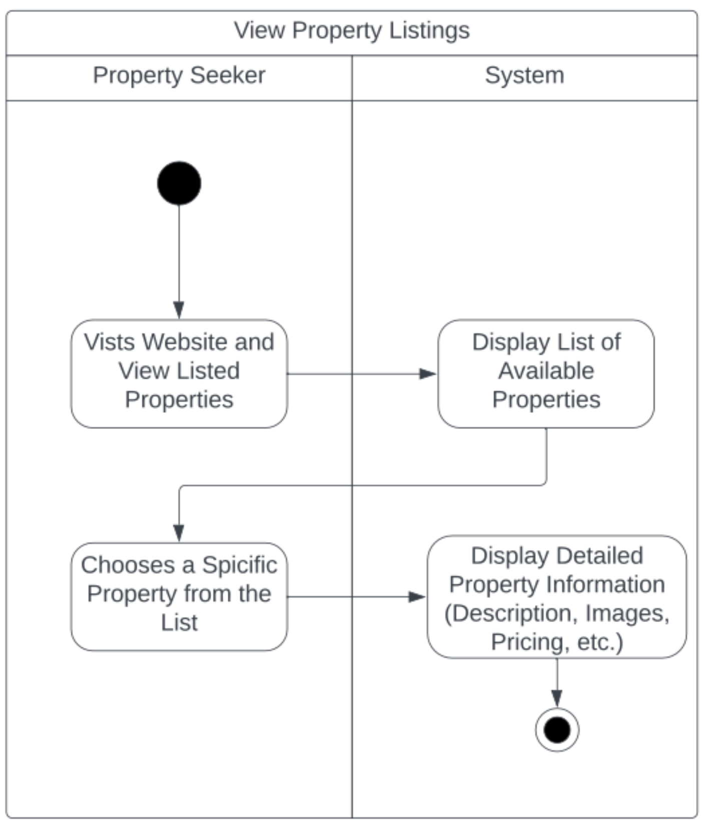
REQ-#	Functional Requirements
REQ-1	The system shall allow the user to log in.
REQ-2	The users shall be able to enter their email addresses.
REQ-3	The users shall be able to enter their password.
REQ-4	The system shall allow the user to send a login request.
REQ-5	The system shall validate the user’s credentials.

REQ-#	Functional Requirements
REQ-6	The system shall take the user to the main page after logging in.
REQ-7	The user shall be able to log out of the system.

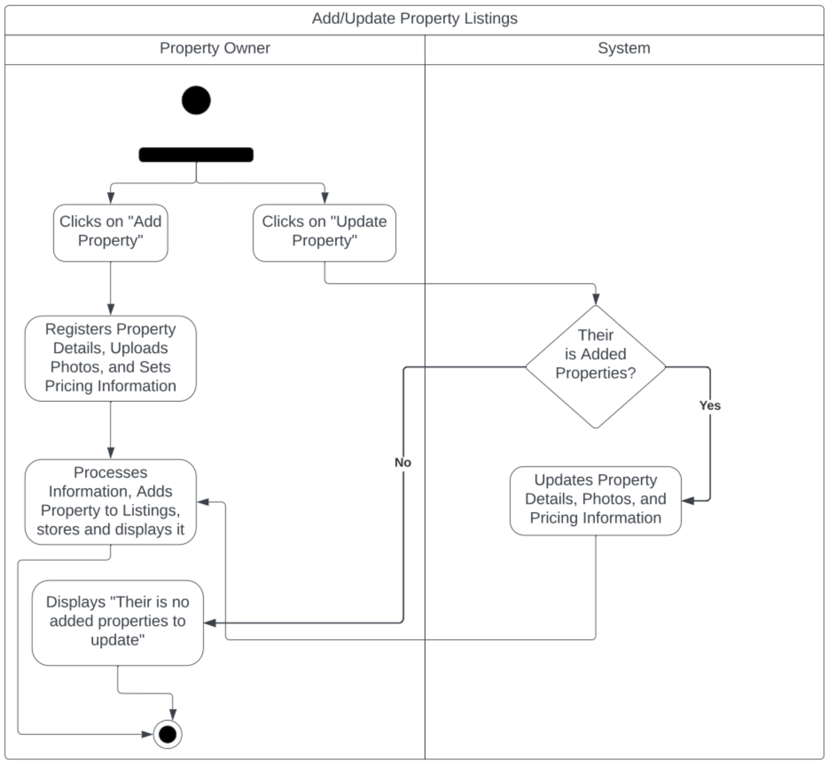
4.3 System Feature 3

4.3.1 Property Listings

a.



b.



Description:

Users will have access to a comprehensive listing of available properties, encompassing villas, apartments, and various housing alternatives, complete with extensive descriptions, photographs, and pricing details. Property owners also have the capability to add and update properties for both rental and sale.

Priority:

High priority, the addition and listing of properties is essential because these features are the core of our website, serving as the primary attraction for both property seekers and owners, making it a valuable and dynamic platform in the real estate market.

4.3.2

1. **Stimulus:** The user visits the website and views the listed properties.

    Response Sequences: The system shows a list of available properties such as villas, apartments, chalets, and other options.

2. **Stimulus:** The user chooses a certain property from the list.

    Response Sequences: The system shows a thorough description, images, and pricing details for the chosen property.

3. Stimulus: logged in property owners can access "Add Property" or "Update Property" where they enter or edit property details, photos, and pricing information.

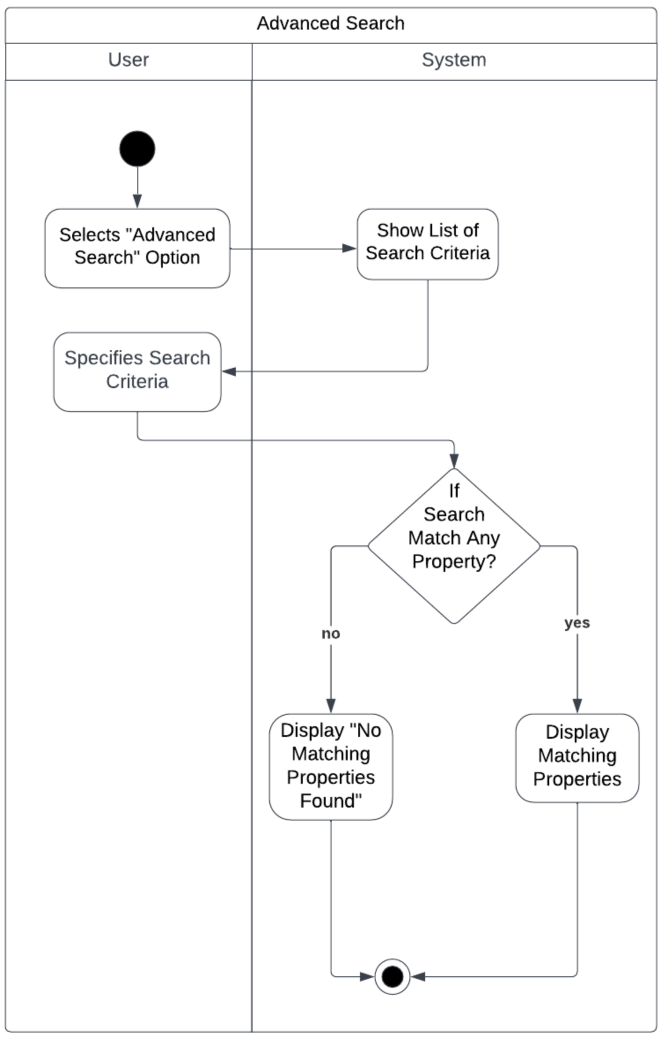
    Response Sequences: The system processes the information, adds the property to the listings,and displays it, or update the property's information.

4.3.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system shall display all available properties within each category.
REQ-2	The system shall display all available properties with detailed descriptions such as, size , number of rooms, and nearby amenities.
REQ-3	The system shall allow users to navigate through available options easily.
REQ-4	The system shall allow property owners to add property for both rental and sale .
REQ-5	The system shall allow property owners to update existing properties.
REQ-6	The system shall allow property owners to input property details.

4.4 System Feature 4

4.4.1 Advanced Search



Description:

A robust search engine allows users to filter properties based on location, type, size, price range, and other relevant criteria to find the properties that match their preferences quickly.

Priority:

High priority, because it empowers users to efficiently narrow down their property search based on specific criteria, ultimately saving them time and ensuring they find listings that closely match their needs and preferences.

4.4.2

1. **Stimulus:** The user selects the "Advanced Search" option from the search menu.

   Response Sequences: The system shows a list of several search criteria, including location, type, size, price range, and additional pertinent criteria.

2. **Stimulus:** The user selects the "Advanced Search" option from the search menu.

   Response Sequences: If the user search does not match any of the property's characteristics, the system displays that no properties match their request.

3. **Stimulus:** The user specifies the search criteria.

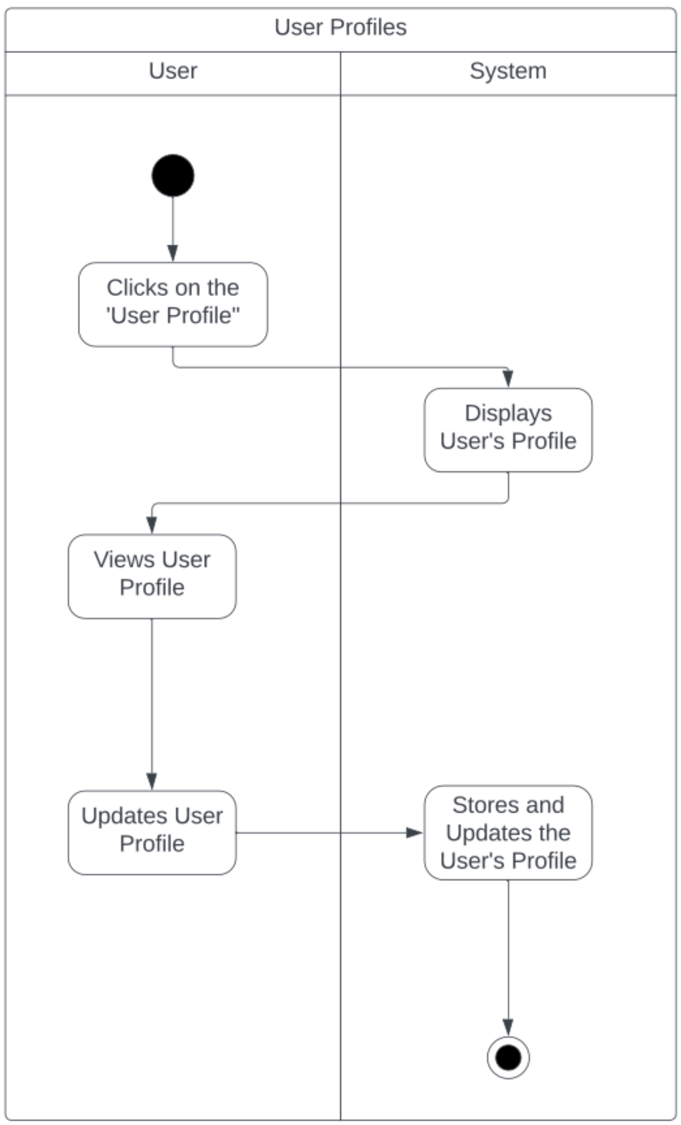
   Response Sequences: The system searches for properties and lists those that meet the user's preferences.

4.4.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	Users shall be able to filter properties depending on their type (villas, apartments, chalets, etc.)
REQ-2	Users shall be able to filter properties depending on their location.
REQ-3	Users shall be able to filter properties depending on their price range.
REQ-4	Users shall be able to filter properties depending on their rating.
REQ-5	Users shall be able to filter properties depending on their desire to rent or buy.
REQ-6	The system must retrieve and display matching properties.

4.5 System Feature 5

4.5.1 User Profiles



Description:

This feature allows users to fill in and edit required fields such as name, city, and phone number, which enables property seekers to save favorite listings, track their search history, and set property alerts for new listings matching their criteria.

Priority:

High Priority, to enable trust-building between property owners and designers.

4.5.2

1. Stimulus: The user clicks on the "User Profile".

Response Sequences: Systems display the user's profile.

1. Stimulus: The user clicks on “Update profile”

Response Sequences: The system gives access to the user's information to update the user profile.

2. Stimulus: Users can view their wishlist listings.

Response Sequences: The system shows the user's wishlist listings.

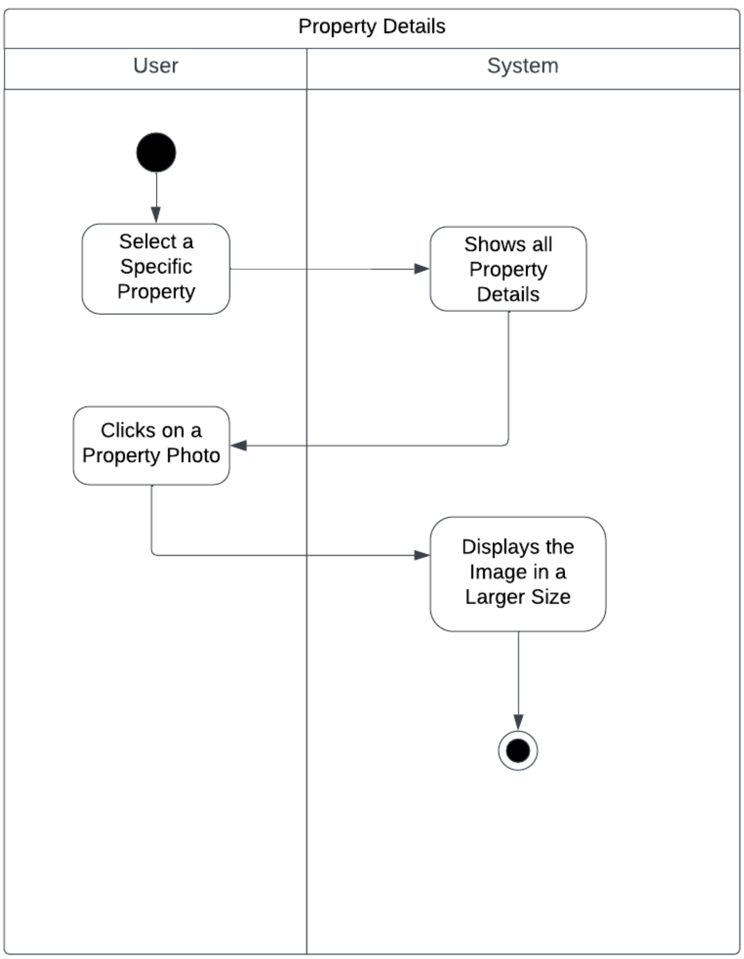
4.5.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system shall allow the user to view their profile.
REQ-2	The system shall allow the user to update their user profile.

REQ-3	The user can view their wishlist listings.

4.6 System Feature 6

4.6.1 Property Details



Description:

The user can see all property information such as nearby schools, hospitals, transportation, and more.

Priority:

High priority, because it provides essential and comprehensive information about a property, enabling potential buyers or renters to make informed decisions with confidence.

4.6.2

1. Stimulus: The user selects the option to view property details.

Response Sequence: The site shows all pertinent details regarding the property, such as the locations of surrounding schools, hospitals, and transit alternatives.

2. Stimulus: The user selects a particular property from the search results or their stored wishlist.

Response Sequences: The system shows specific details about the property, such as the address, size, cost, and amenities.

3. Stimulus: The user clicks on a property photo to view it in its entirety.

Response Sequences: The system displays the image in a larger size for easier viewing.

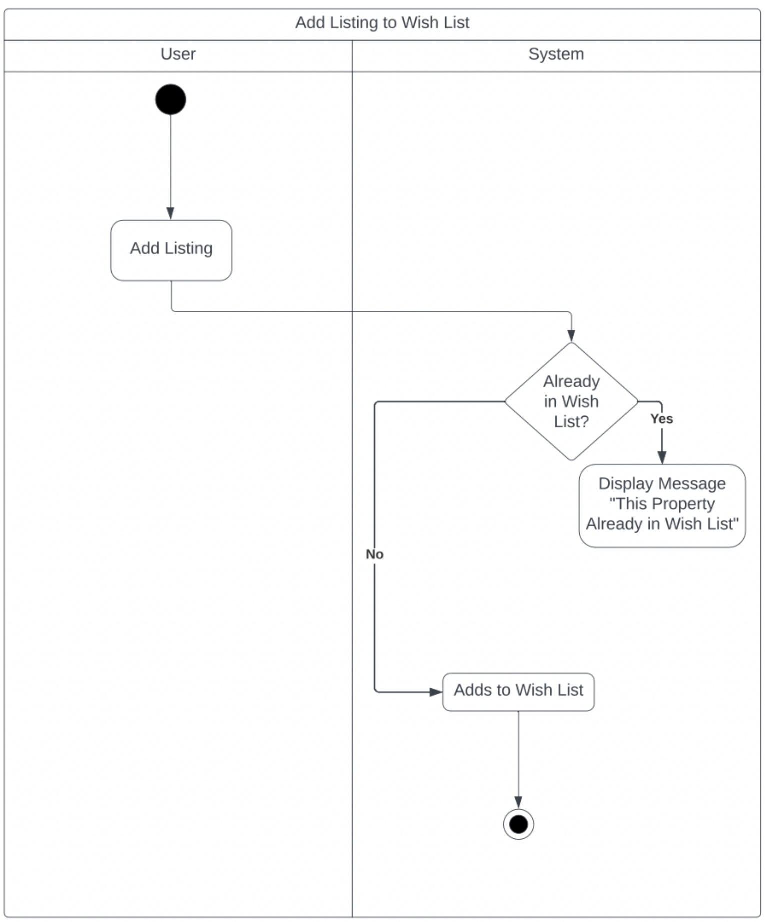
4.6.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system shall allow users to view property details.
REQ-2	The system shall provide an interface for users to view property details.
REQ-3	The system shall allow the user to enlarge the property photos.

4.7 System Feature 7

4.7.1 Wish List

a.



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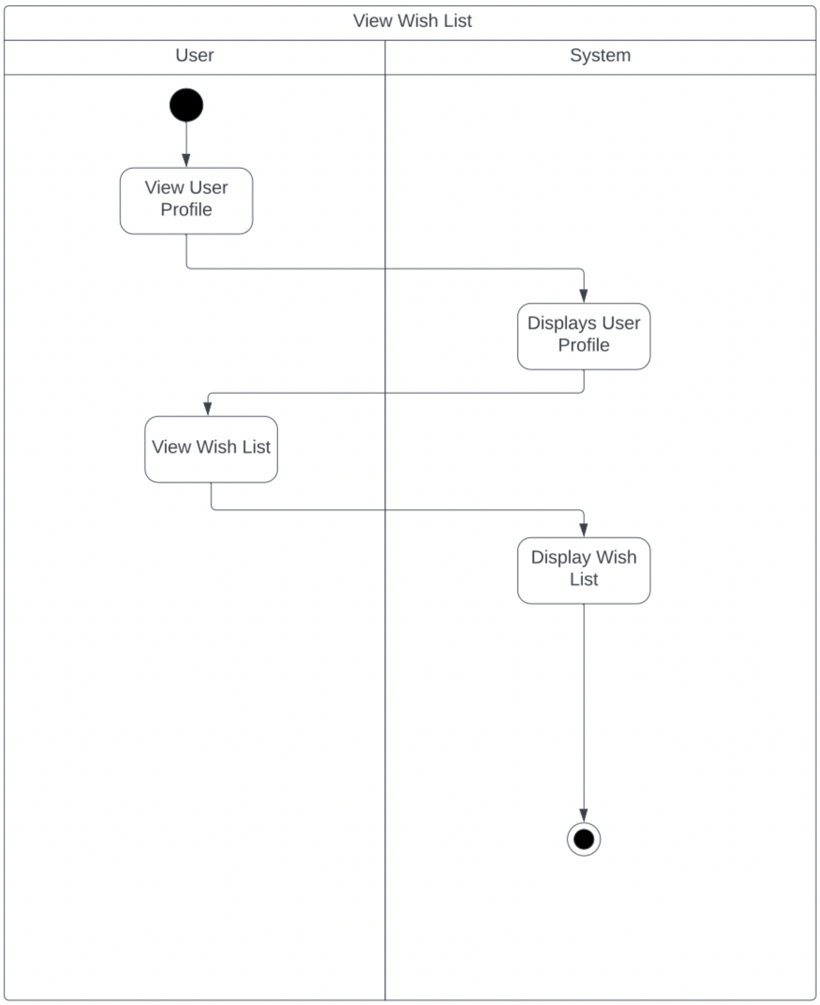


Description:

The HomeKey website provides a wish list where the registered user will be able to save the liked property or interior design to the favorites list, as it makes it easier for them to find it later on.

Priority:

Medium priority.

4.7.2

1. Stimulus: The user adds a property listing or an interior designer to their wish list.

Response Sequence: The system must add the selected property or interior designer to their wish list.

2. Stimulus: The user views their wish list.

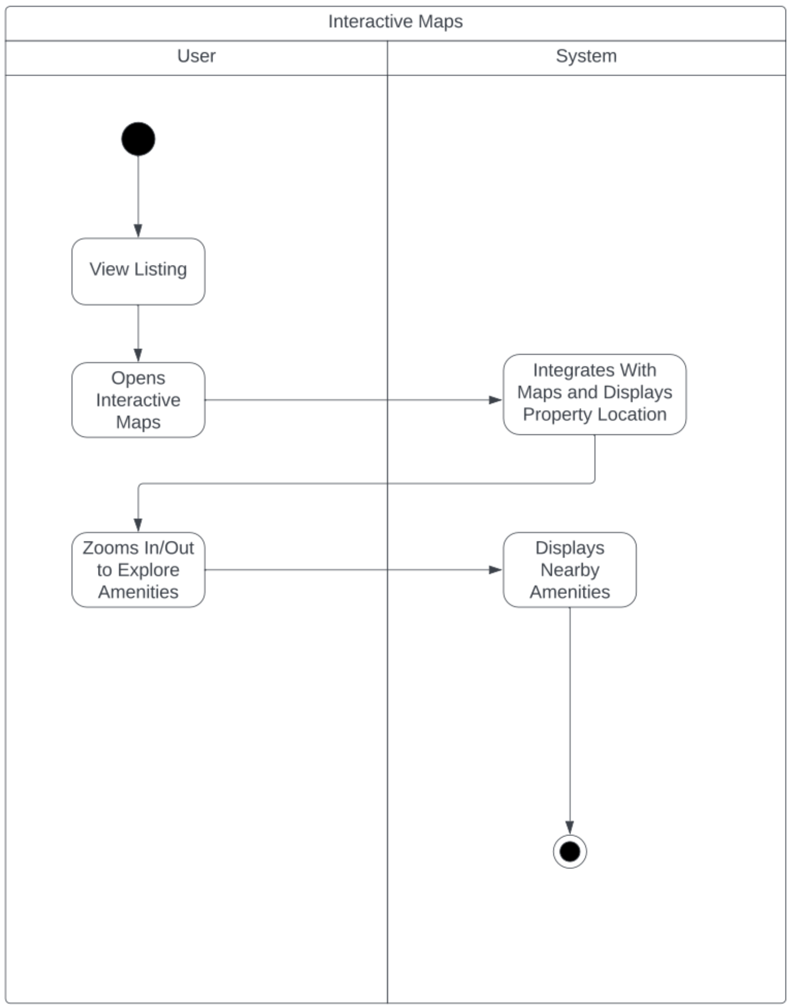
Response Sequence: The system displays the wish list to the user.

4.7.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The user must be logged in to the website.
REQ-2	The system shall allow the user to add a property or an interior designer to the wishlist.
REQ-3	The system shall allow registered users to access their wish list.
REQ-4	The system shall display the user's saved wish list for properties or interior designers.

4.8 System Feature 8

4.8.1 Interactive Maps


Interactive Maps
User
System
View Listing
Opens Interactive Maps
Integrates With Maps and Displays Property Location
Zooms In/Out to Explore Amenities
Displays Nearby Amenities

Description:

Integration with maps to show property locations and nearby amenities, schools, parks, and public transportation.

Priority:

Medium priority.

4.8.2

1. Stimulus: The user opens an interactive map on the website.

Response Sequences: The system integrates with maps and displays the location of the property on a map interface.

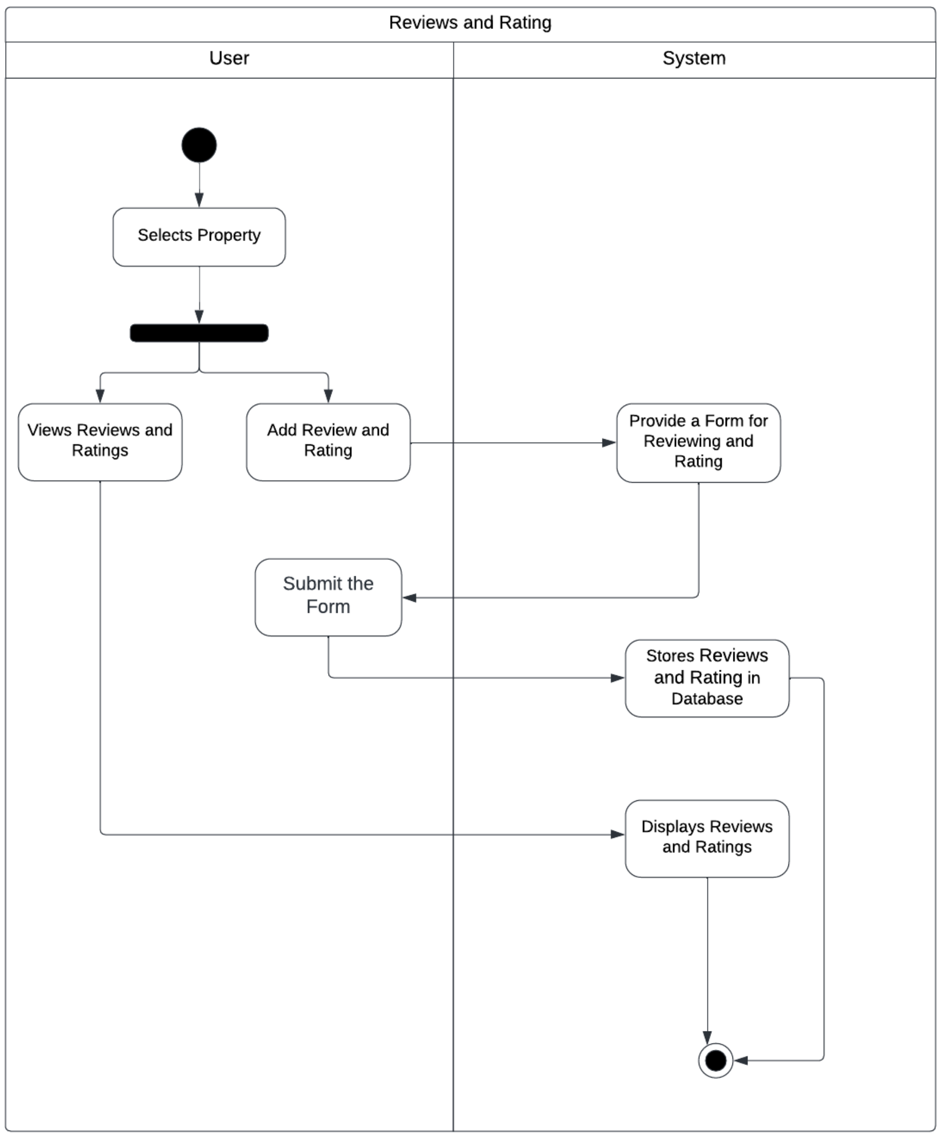
2. Stimulus: Users can zoom in/out and pan around to explore nearby amenities such as schools, and public transportation.

4.8.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system shall be able to integrate with popular mapping programs such as (Google Maps) to display property locations accurately.
REQ-2	The system shall provide a visual representation of property locations.
REQ-3	The system shall display nearby amenities.
REQ-4	Users shall be able to easily identify the position of each property on the map.

4.9 System Feature 9

4.9.1 Reviews and Ratings



Description:

Allow users to leave reviews and ratings for properties they have visited, helping others make informed decisions.

Priority:

Medium priority.

4.9.2

1. **Stimulus:** The user selects a property they have visited or liked.

Response Sequences: The System shows any reviews and ratings that have already been submitted for that property.

2. **Stimulus:** The user submits their own review and rating.

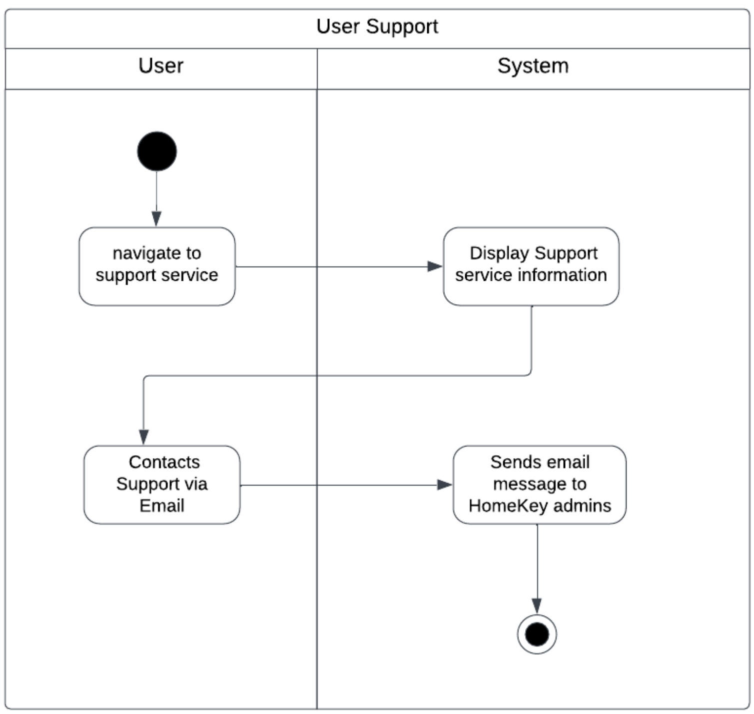
Response Sequences: The system verifies that the review or rating was successfully submitted.

4.9.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system shall allow registered users to review and rate.
REQ-2	The system shall provide a form or interface for users to add their reviews and rating scores.
REQ-3	The system shall allow the user to view reviews and ratings written by other users.
REQ-4	The system shall store submitted reviews and ratings.

4.10 System Feature 10

4.10.1 User Support



Description:

Offering customer support service via email address to assist users with inquiries or issues.

Priority:

High priority, because it directly impacts the overall user experience, ensuring that users feel valued and supported when they encounter questions or issues, which in turn fosters trust and loyalty.

4.10.2

1. **Stimulus:** User navigates to support services

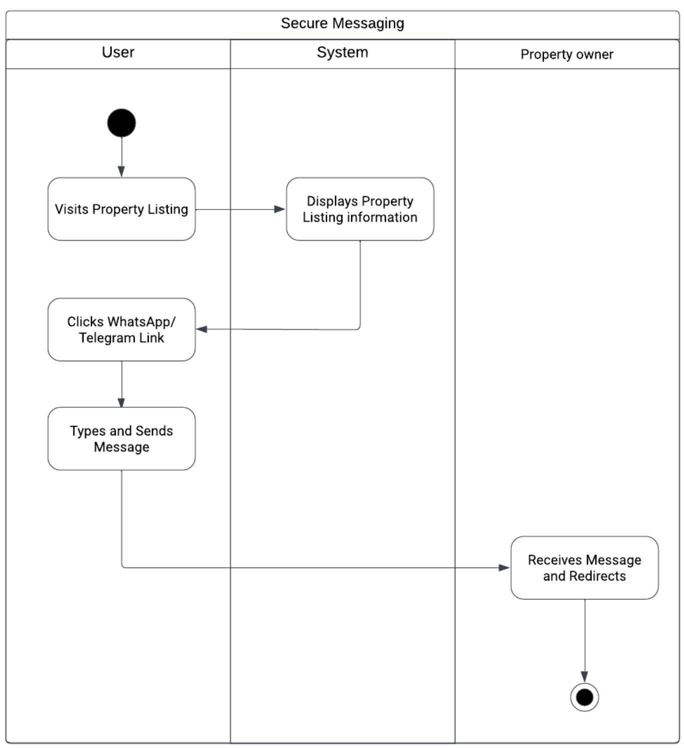
Response sequence: The system displays support services information.

4.10.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system shall provide an email address for support services.
REQ-2	The system must be able to support and handle user inquiries effectively.
REQ-3	The systems must be able to provide quick responses and solutions to user questions or problems.

4.11 System Feature 11

4.11.1 Secure Messaging



Description:

WhatsApp or Telegram links are used as a communication channel for property seekers, property owners, and real estate agents to ensure a seamless and secure user experience.

Priority:

High priority, because it safeguards user privacy and personal data, ensuring that their conversations remain confidential and protected from potential threats or intrusions, ultimately fostering trust and peace of mind among users.

4.11.2

1. **Stimulus:** User visits HomeKey website and finds a property they are interested in.

Response Sequence: The system displays the WhatsApp or Telegram link provided for that specific property to communicate with the property owner.

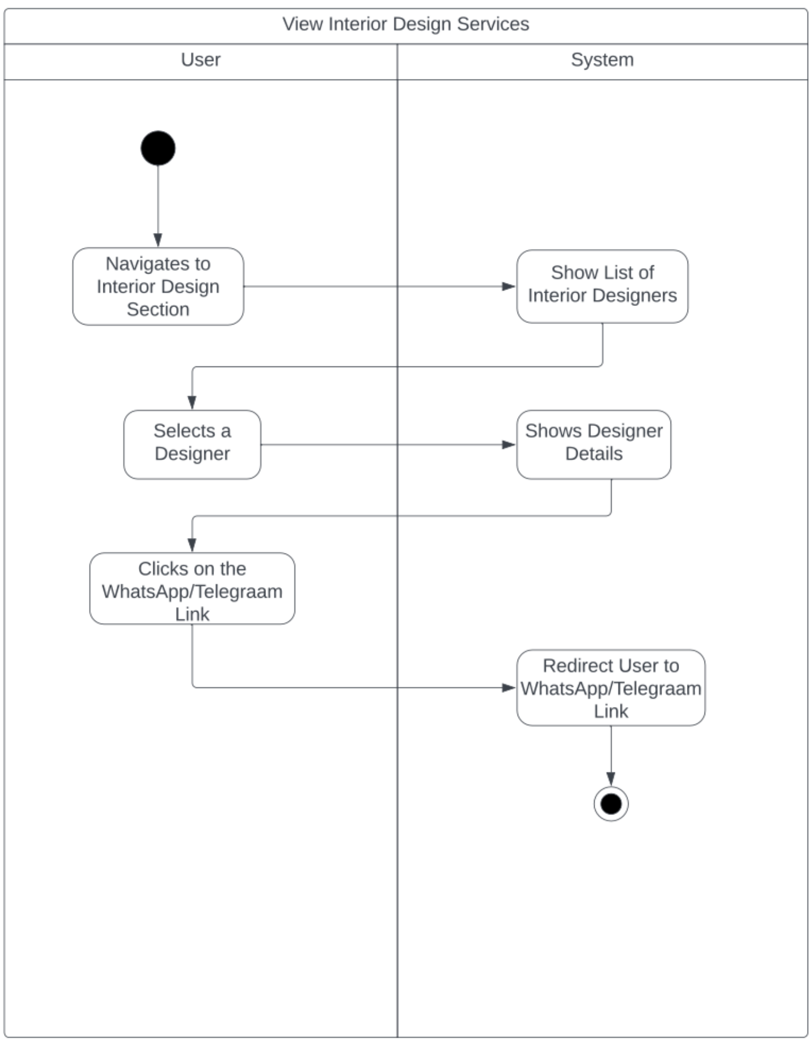
4.11.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system shall allow the property owners and interior designers to add their WhatsApp or Telegram link in the contact information.
REQ-2	The system shall redirect the property seeker to WhatsApp or Telegram chat.

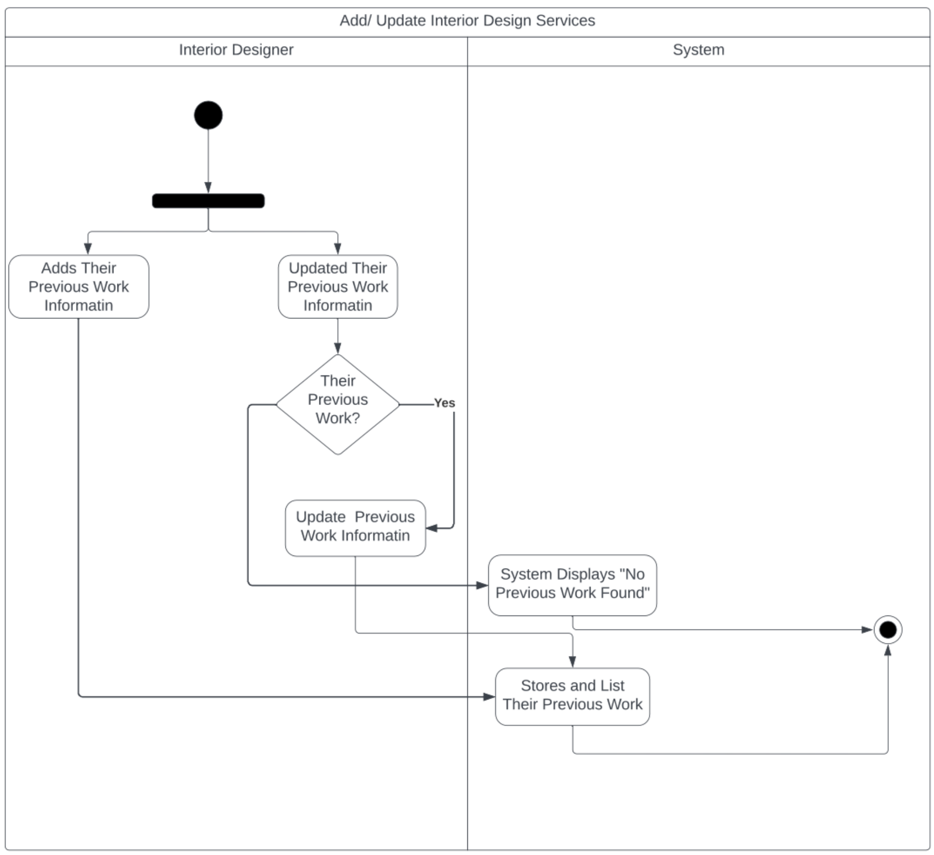
4.12 System Feature 12

4.12.1 Interior Design Services

a.



b.



Description:

We offer a dedicated section that allows users to get in touch with interior designers. This service is designed to assist those who either wish to streamline their interior design process or may lack the expertise to decorate their space effectively. It's a valuable resource for anyone looking to enhance their home's aesthetic appeal.

Priority:

High priority, since it aligns with the unique preferences and demands expressed by our users through the Google Form survey. As it offers our users a unique and valuable dimension that enhances their property exploration by providing personalized design solutions and creating a memorable and distinctive user experience.

4.12.2

1. Stimulus: The user navigate to the interior designs section

Response sequence: The system shows a list of interior designers

2. Stimulus: The user selects the designer he/she likes .

Response sequence: The system shows details about the designer in terms of previous work.

3. Stimulus: The user requests to communicate with the interior designer.

Response sequence: The system displays a Whatsapp or Telegram link to communicate.

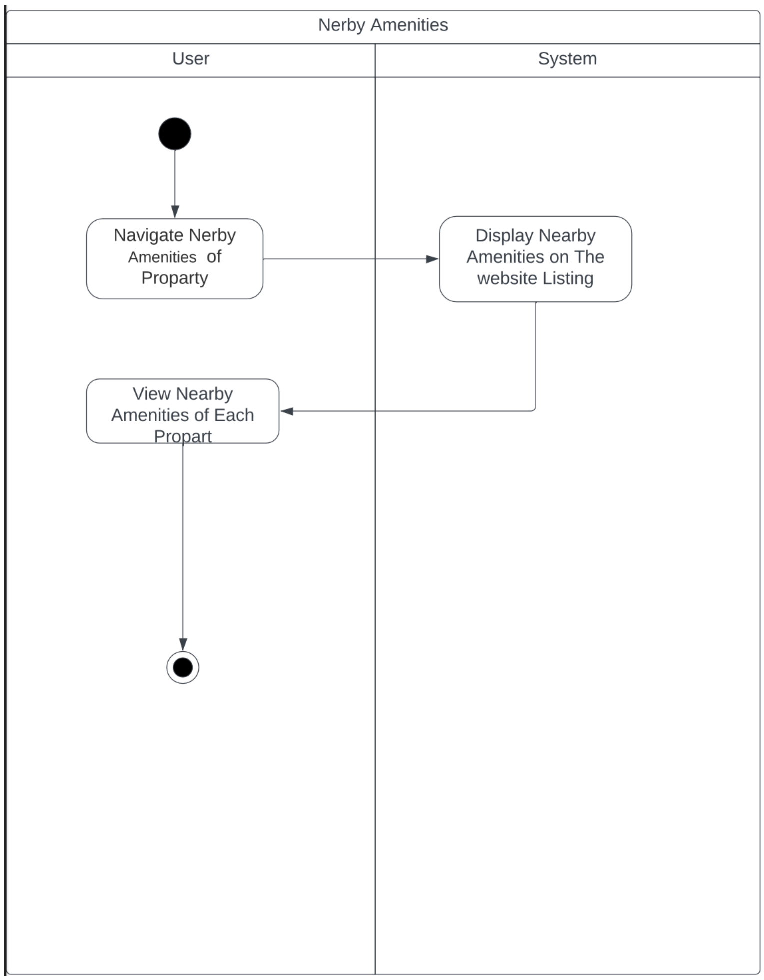
4.12.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system provides a dedicated section on the website for interior design services.
REQ-2	The system shall allow the interior designer to add their previous work information.
REQ-3	The system shall allow the interior designer to update their previous work information.
REQ-4	The system shall allow users to view interior designer's profiles.
REQ-5	The system shall allow users to see the previous work of interior designers.
REQ-6	The system shall allow registered users to communicate with interior designers.

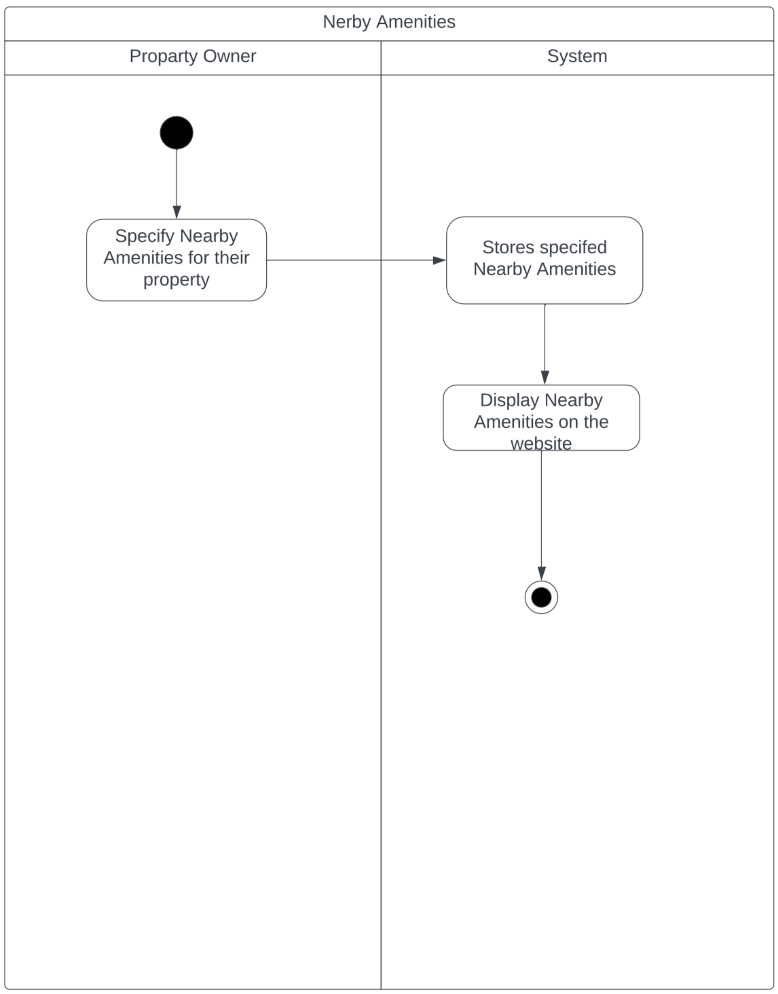
4.13 System Feature 13

4.13.1 Nearby Amenities

a.



b.



Nerby Amenities
Proparty Owner
System
Specify Nearby Amenities for their property
Stores specifed Nearby Amenities
Display Nearby Amenities on the website

Description:

Our website provides users with the ability to view nearby amenities, enabling them to gain a clearer understanding of the vicinity and determine whether a location aligns with their preferences and requirements.

Priority:

Incorporating the "Nearby Amenities" feature as a **high priority** aligns with our users' preferences and needs, as highlighted by their responses in the Google Form survey. By offering this feature, we prioritize user satisfaction and provide valuable information that assists them in making informed property decisions based on their desired amenities and lifestyle.

4.13.2

1. **Stimulus:** Users view nearby amenities

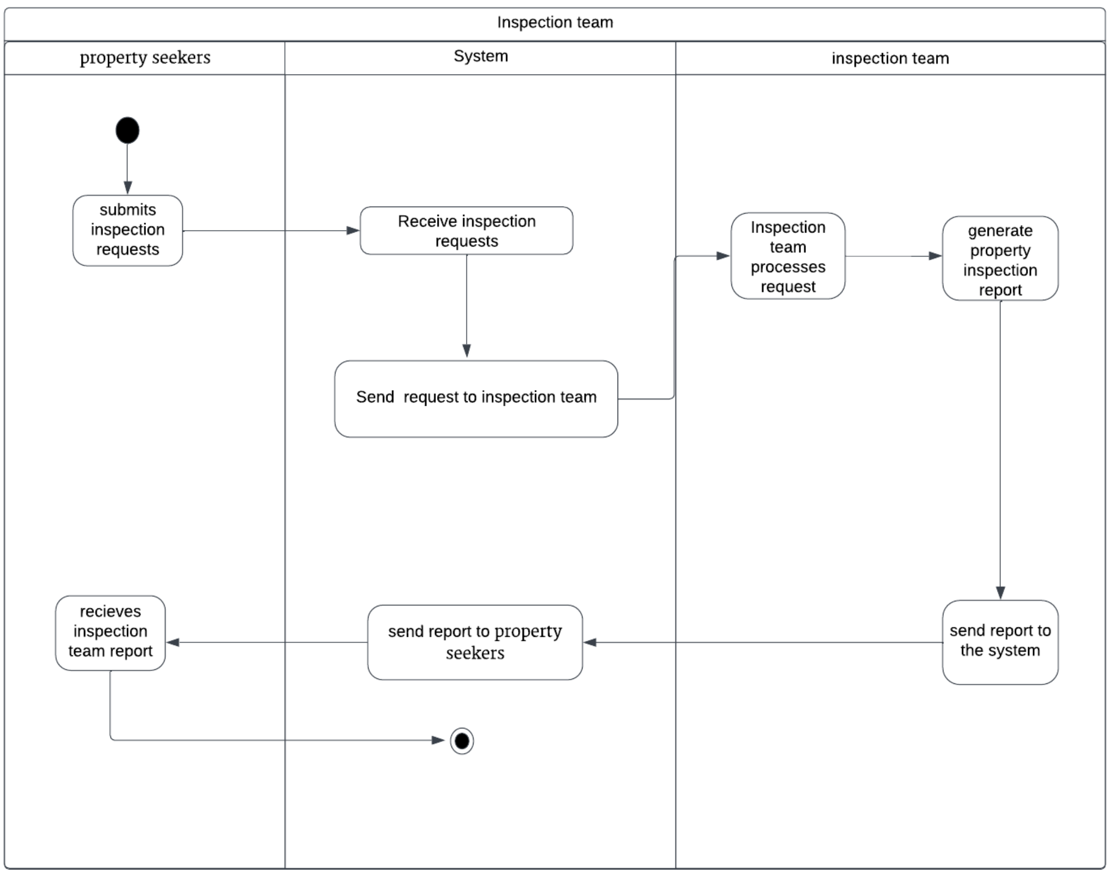
Response sequence: The system displays the nearby amenities.

4.13.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system shall allow property owners to specify nearby amenities for their properties.
REQ-2	The system shall allow users to view nearby amenities of each property.

4.14 System Feature 14

4.14.1 Inspection team



Description:

Our website allows the buyer or tenant to request an inspection team to verify the presence of any defect or problem in the property. We receive the customer's request and we are the ones who undertake the inspection team's request to go to the property. After completion, a report is submitted to us about all the required defects. After that, the report is sent to the customer.

Priority:

High priority, because it distinguishes our website from competitors, offering our users a distinctive and valuable service that bolsters their confidence in property decisions and sets us apart as a trusted and comprehensive platform for their property needs.

4.14.2

1. **Stimulus:** property seekers, clicks on the inspection team service button.

Response Sequence: The system displays the Inspection Team service.

2. **Stimulus:** The property seekers initiates the property inspection request by providing his or her contact detail (email address).

Response Sequence: The system sends the request to the inspection team

3. **Stimulus:** the inspection team reports all defects and problems recommended for treatment and submit it to the system

Response sequence: The system sends the inspection report to the user via his email address.

4.14.3 Functional Requirements

REQ-#	Functional Requirements
REQ-1	The system shall allow property seekers to request property inspection.
REQ-2	The user should provide their email addresses.
REQ-3	The system should send the request to the inspection team.
REQ-4	The inspection team should submit the inspection report to the system.
REQ-5	The system shall send the inspection report to the property seekers.

5. Other Nonfunctional Requirements

5.1 Security Requirements

SCR-1: The system shall require user authentication for access.

SCR-2: The system shall ensure that no two accounts are registered with the same email.

SCR-3: The system shall ensure that no two accounts are registered with the same phone number.

SCR-4: The system shall require passwords to be at least 8 characters long.

5.2 Performance Requirements

PR-1: The system shall ensure that the user interface is responsive, and it shall promptly and accurately apply users action in 3 seconds.

5.3 Software Quality Attributes

QA-1: Availability:
The system shall be available to users 97% of the time.

QA-2: Usability:
The system shall allow the property owner to add a new property in 12 minutes.

QA-3: Usability:
The system shall allow users to complete the signup process in 5 minutes.

QA-4: Maintainability:
The system's design shall facilitate effective problem diagnosis and troubleshooting by offering clear error messages.

5.2 Business Rules.

BR-1: The system will only accept bookings within Riyadh for the first release of the website.

6. Other Requirements

OR-1: HomeKey shall be released in English only for the first release.

Appendix A: Glossary

1- **Property Owner:** Occasional users who list their properties for rent, book and buy.
2- **Property Seeker:** Regular users visit the website to find properties for rent, booking or buying.
3- **Interior Designer:** Occasional users who provide interior design services.
4- **Inspection Team:** The individuals tasked with evaluating the property and submitting their inspection report to our system.
5- **UI:** User Interface
6- **Admin or Team:** The person who is responsible for managing the website.

Appendix B: Focus Group Scripts and Survey Results

Focus Group Google Form Link: https://forms.gle/AWvHKGp7pJpH8UjZA

What's You're Name?

5 responses

- Alanoud Abdullah
- Sara Alshareef
- Noura Khaled AlZahrani
- Remas AlMuhanna
- Amasi Nawaf

How Old Are You?

5 responses

- 25
- 28
- 21
- 26
- 22

How often do you use property booking or rental apps in Saudi Arabia?

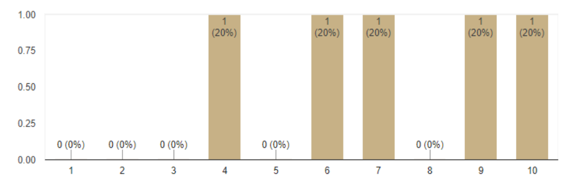
5 responses

- around twice a week total of 8 times a month
- always
- one - twice a month
- from 2-3 times a week
- ALOT

Rate your satisfaction with current apps like Airbnb, Gathern, and Aqqar in Saudi Arabia.

Copy

5 responses



Share your experience using Airbnb, Gathern, and Aqqar.

5 responses

- nice but not the best tbh
- i like using them but they need to add additional features
- pleasant good experience
- not the best but very needed
- good, clear, easy to use, i see it very comfortable

What feature do you find most convenient when using these apps?

5 responses

- property listing and their diversity
- the way they list and have a lot of places and choices from chalets to villas to apartments
- easy search
- variety of properties for summer winter work living and more
- that i can see all properties

What feature do you dislike the most in these apps? Why?

5 responses

- i cant view what is near me when i choose a place to book i have to open google maps
- when they don't add property owner number so i have to wait for their response (message) if im in a hurry i would really prefer calling them (remember the real life scenario of the studio i booked when i was in a hurry)
- very slow costumer service
- when the application or website is slow
- nothing

What feature do you like the most in these apps? Why?

5 responses

- the ease of booking
- the filtering part
- even if its my first time using the app ill know how to search and choose what i want to book. in short, the ease of booking is what i like the most
- easy to use I see using them is very convenient
- most of them

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What features or functions would you like to see in property booking or rental apps?

5 responses

- i want to see what i have near the place ill book
- calling the owner for faster response or emergency in the place i booked
- i have no feature to add but would like to make the costumer service better
- people to check and make sure that things are safe and in good condition, find and fix any problems or hidden issues to keep me safe and happy.
- i think what they have is good but we can never say no to good additional features

What factors are most important to you when choosing such apps?

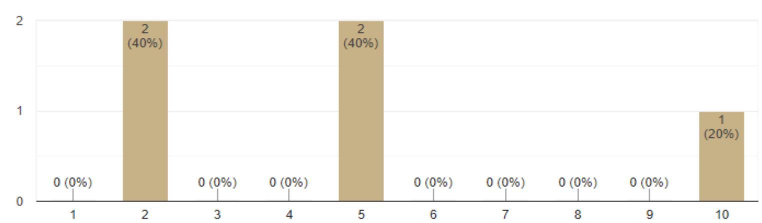
5 responses

- location شمال الرياض
- safety of the app ,, is it safe? then ill totally use it
- that they have a section for reviews and feedbacks so i can read them before booking
- easy communication with owner
- the property owner responses سرعة الرد و الاسلوب

How important is it for you to have a multilingual interface in property apps, including Arabic?

Copy

5 responses



What role do local recommendations play in your decision to book or rent a property?

5 responses

- people's rating yes and i trust saudi people feedbacks
- honestly nothing
- nothing
- for me, no
- يهمني اقرء تعليقاتهم و اشوف تقييماتهم لا اكثر

Home Key

If you could remove certain features, which ones and why?

5 responses

- none ! i want to add haha
- not deleting i want to add
- nothing
- none i assume all is important
- all is necessary for me

Are there specific cultural or local requirements you believe these apps should address for the Saudi market?

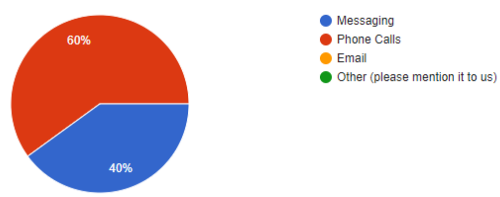
5 responses

- nothing
- no..
- No
- no

What is your preferred method of communicating with property owners or renters? For example, messaging, phone calls, or other options?

Copy

5 responses



Lets assume you will develop an application for property renting, what specific services you'll add?

5 responses

- more info about location
- phone number , location even though it wrong
- automatic costumer service or maybe a faster one
- people who check before long lasting renting or booking
- better pictures and CLEAR videos for the properties

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PRINCE SULTAN UNIVERSITY

How do you believe property apps can improve the user experience for people in Saudi Arabia?

5 responses

- i think they are great except for the location (nearby)
- add arabic
- i think all is good, saudi people are not like before
- offers on properties
- احس اننا شعب يعجبنا كل شي و سريعين الاقتناع

What specific information are you interested in regarding the property OWNER?

5 responses

- nationality cuz i prefer to rent from saudi people tbh
- if his phone number is added, goof reviews
- nothing special maybe their nationality sometimes?
- honestly i don't care about him the property is all i care about .. but i want him/her to reply fast thats it
- name and properties they have

What details do you want to gather about the property ITSELF?

5 responses

- WHAT IS NEAR , المساحة لان الصور مب كل شي
- district and price ofcourse
- size and what floor
- main road and district price
- size location district price

How would you prefer to search for a particular property or specific property features?

5 responses

- by filtering location and sometimes price depends on the budget haha
- manually type what i want to book because im always in a hurry when looking for a place to book
- i can search regularly or filter it depends
- based on price
- filter becuase its easier

Home Key

What personal account information would you like to include and feel at ease adding to your account?

5 responses

- phone num , email
- name, email, phone, age i dont mind all
- name, email, phone, id,
- name email number sex
- name number email

And That's A Wrap!

Is there anything else you would like to contribute?

Please feel free to share your thoughts in the space provided below, and don't hesitate to share them now.

We are here to listen..

Thank you for your valuable input and your time!

5 responses

- ياحبيلكم يعطيكم العافية سريعين و لطيفين
- thankyou ladies!!
- thx!
- Thank u for your time and effort best of luck
- i wish all of you the very best luck الله يوفقكم و عقبال نشوفه انجح ابلكيشن بالدنيا

Appendix C: Workshop Scripts and Survey Result

Workshop Google Form link: https://forms.gle/scYWdG4CMg87roKe6

My name is:

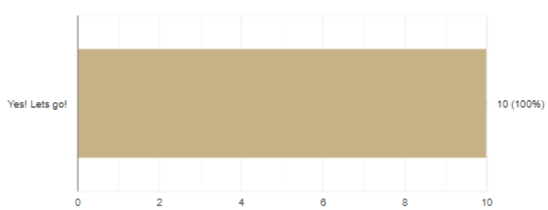
10 responses

- salwa aldarwesih
- Deema Shihadeh
- atheer alsaid
- sara alsaid
- layal alkhodair
- Alanoud Abdullah
- Sara Alshareef
- Noura Khaled AlZahrani
- Remas AlMuhanna

I have a clear understanding of the system's scope and functionality and will provide effective answers and priorities.

Copy

10 responses

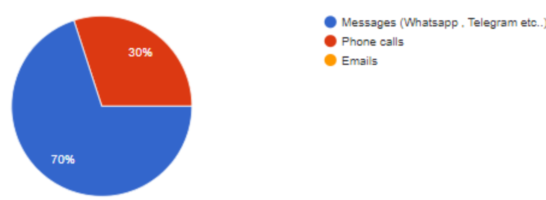


Features and Prioritization

What is your preferred method of communicating with property owners or renters?

Copy

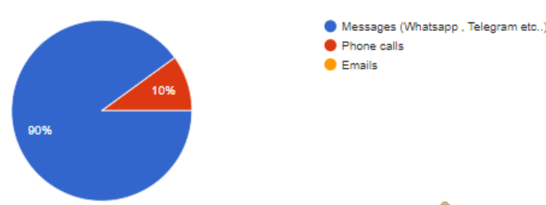
10 responses



What is your preferred method of communicating with Interior designers?

Copy

10 responses



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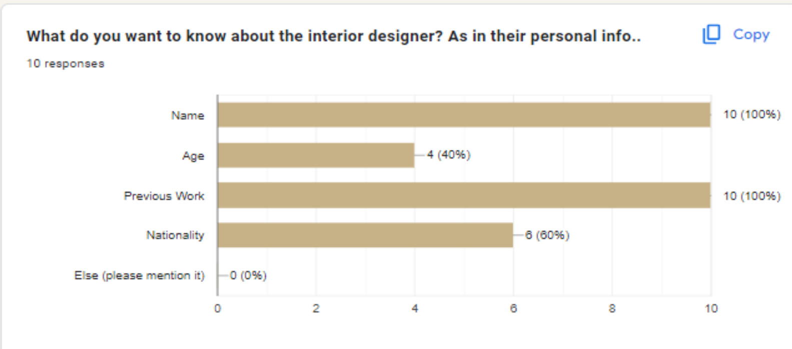
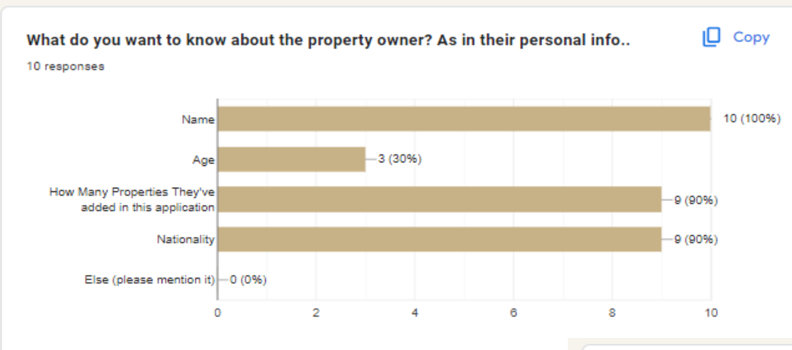
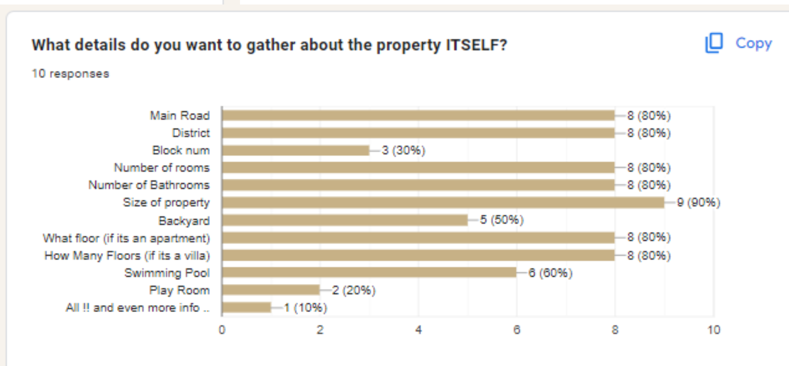


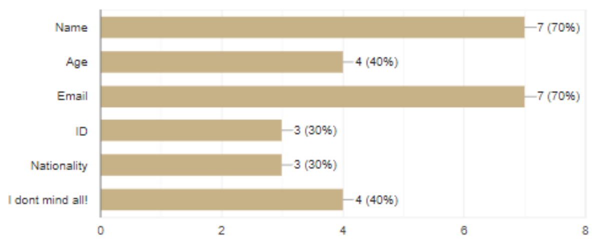




Home Key

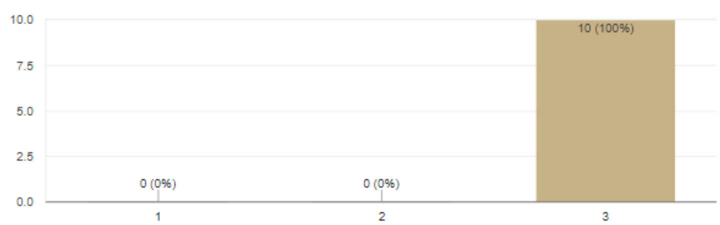
What personal account information would you like to include and feel at ease adding to your account?

10 responses



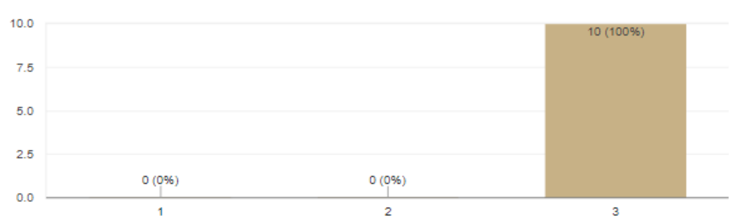
On a scale of importance from 1 to 3, How important is the Interior Design Services?

10 responses



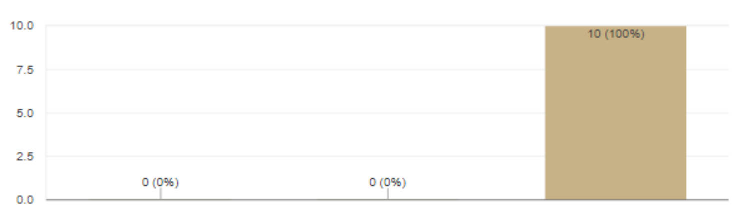
On a scale of importance from 1 to 3, How important is the Inspection team Services?

10 responses



On a scale of importance from 1 to 3, how significant is it for you to have access to information about nearby amenities?

10 responses



Rank these features based on your preferences: Information about nearby amenities, Interior designers, Inspection Team.

10 responses

- 1-nearby amenities, 2-Interior designers, 3-Inspection Team.
- First, Nearby Amenities
  Second, Interior Designers
  Third, Inspection Team
- 1- nearby amenities 2- interior designers 3-inspection team
- 1 nearby amenities 2 designers 3 inspection team
- 1/ nearby amenities - 2/ interior designers 3/ inspection team.
- 1- Interior designers 2- amenties 3-inspection
- Inspection Team then id then nearby
- الله لا يحير مسلم كبركوهم بعيننا بس احس الاشياء القريبه مهمه جدا من ثم الديكور بعدين انسبكشن

And That's A Wrap!

Please feel free to share your thoughts in the space provided below.

Thank you for your valuable input and your time!

6 responses

- Adding a favorite section to be able to save the properties that I liked
  يعطيكم العافيه
- Favorite / wishlist
- يعطيكم العافيه
- thanks ladies goodluck
- شكرا لكم
- goodluck!

SDS

SOFTWARE DESIGN SPECIFICATION

1. Introduction

1.1 Purpose:

This document, the Software Design Specification (SDS), lays out the plan for building the HomeKey website. It gives clear instructions on how the website should be constructed to meet specific requirements. This document is essential for the project team as it provides detailed information needed to achieve the goals of the HomeKey website. It will be an important guide, helping us create the website exactly as the requirements state.

2. System Architecture

2.1 Architectural Style

The three-tier architecture is a highly refined and carefully organized framework that serves as the foundation for various software applications. It meticulously separates these applications into three specific layers, each equipped with its own unique purpose and self-reliant capabilities. This meticulous division leads to the creation of a system that is not only remarkably well-organized but also effortlessly adaptable to growth and expansion. This means that as the needs of the application evolve, each layer can be adjusted independently, allowing for seamless enhancements and adaptations.To put it concisely, the selected architecture for HomeKey is the three-tier architecture, and there are numerous compelling reasons for this choice, which will be elaborated upon below.

2.2 Layers / Tiers

Presentation Layer:
In the HomeKey system, the presentation layer serves as the user's gateway to the platform. It is the part of the system that users directly interact with.
This layer is responsible for displaying the website interface that users see when they visit HomeKey. It handles user inputs, such as search queries, property selections, and user account interactions. Additionally, the presentation layer ensures that the user experience is intuitive and seamless. It provides features like advanced search options, user profiles and more. These elements make it easy for users to find and interact with properties.

Application / Business logic layer:
Business logic layer acting as the intelligence behind the platform's functionalities. This layer is responsible for processing data and executing operations. It manages tasks such as property listings, reviews and ratings, secure messaging, and interior design services. The business logic layer orchestrates the logic that drives the behavior of the platform. For example, it processes what users search to match with relevant property listings and handles a lot more.

Data Storage Layer:
The data storage layer is like the digital filing cabinet of HomeKey. It securely stores and retrieves information related to properties, user accounts, reviews, and more. This layer manages databases containing details about various properties, user profiles, and reviews. It ensures that data is organized, accessible, and protected. The data storage layer also enables features like property details and nearby amenities. It provides the necessary information to users, allowing them to make informed decisions about renting or buying properties.

To summarize, the three-tier architecture of HomeKey effectively separates and delegates specific tasks to each layer. The presentation layer facilitates user interaction, the business logic layer handles the core functionalities and operations, and the data storage layer manages the storage and retrieval of essential information. This architecture ensures that HomeKey operates efficiently and provides a seamless experience for users seeking to book, rent, or buy properties in Saudi Arabia.

2.3 The Rationale for Choice

The rationale behind choosing the three-tier architecture for HomeKey:

1. **Quality Attributes:**
   We selected the three-tier architecture because it helps us meet our quality goals. This means our website will meet its nonfunctional requirements successfully.

2.4 Quality Attributes of interest

Availability:

The architecture separates the system into distinct layers: presentation, application, and data tiers. This separation of layers ensures that issues in one layer are less likely to affect the others. This minimizes downtime due to localized problems.

Usability:

The system is divided into three distinct layers: presentation, application, and data tiers. This division assigns specific roles to each layer, simplifying the system's management and comprehension. It also enables designers and developers to focus on creating an aesthetically pleasing user interface.

Maintainability:

The 3-tier architecture enhances maintainability by isolating each layer, ensuring that modifications made to one layer doesn't impact other layers, thus facilitating efficient updates and maintenance.

To sum up, we chose the three-tier architecture for our website. It provides advantages in three areas. It ensures high availability by isolating layers, reducing downtime. It enhances usability by simplifying system management and enabling a focus on a pleasing user interface. Lastly, it promotes easy maintenance by isolating layers, allowing for efficient updates.

2.5 Architectural Views | 4+1 Model

User View - Use Case Diagram

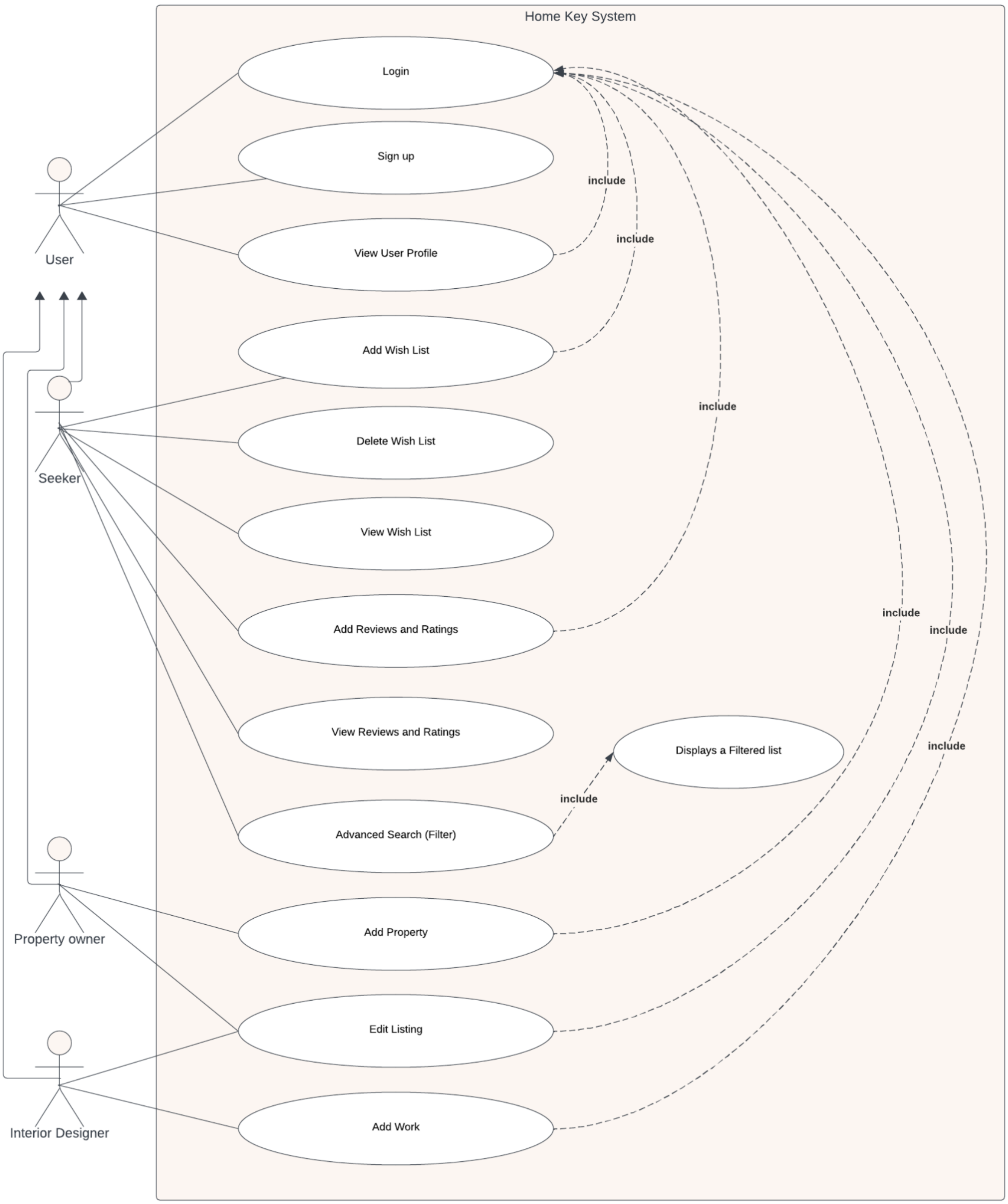
Logical View - Data Model Diagram

Development View - Component Diagram

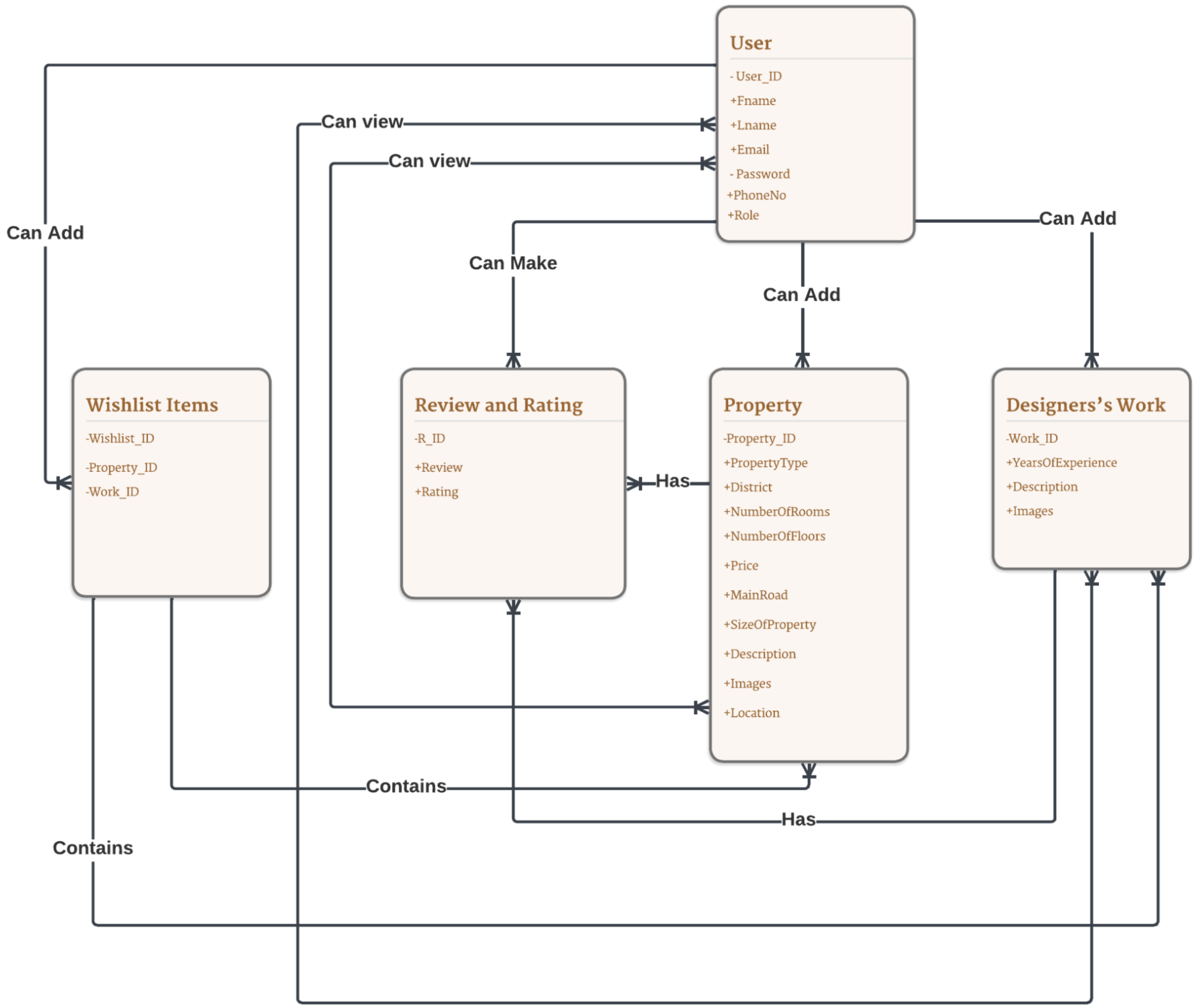
Process View - Activity Diagram

Physical View - Deployment Diagram

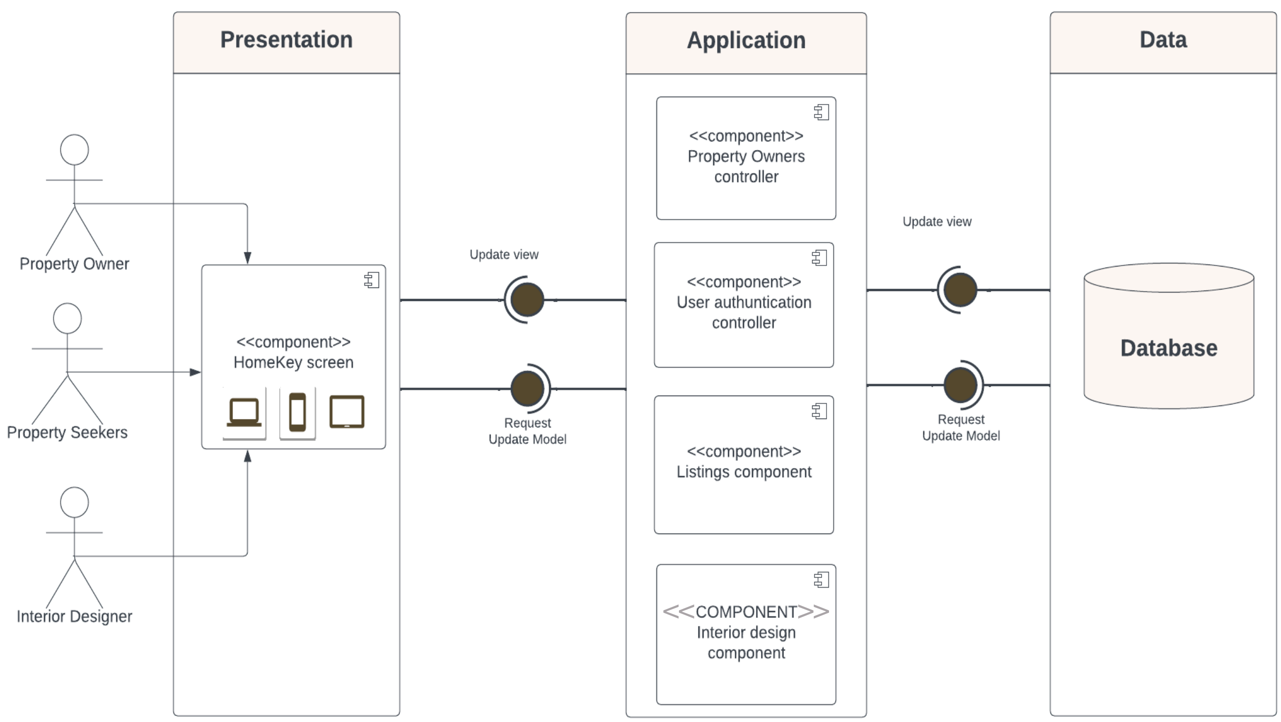
User View - Use Case Diagram



Logical View – Data Model Diagram



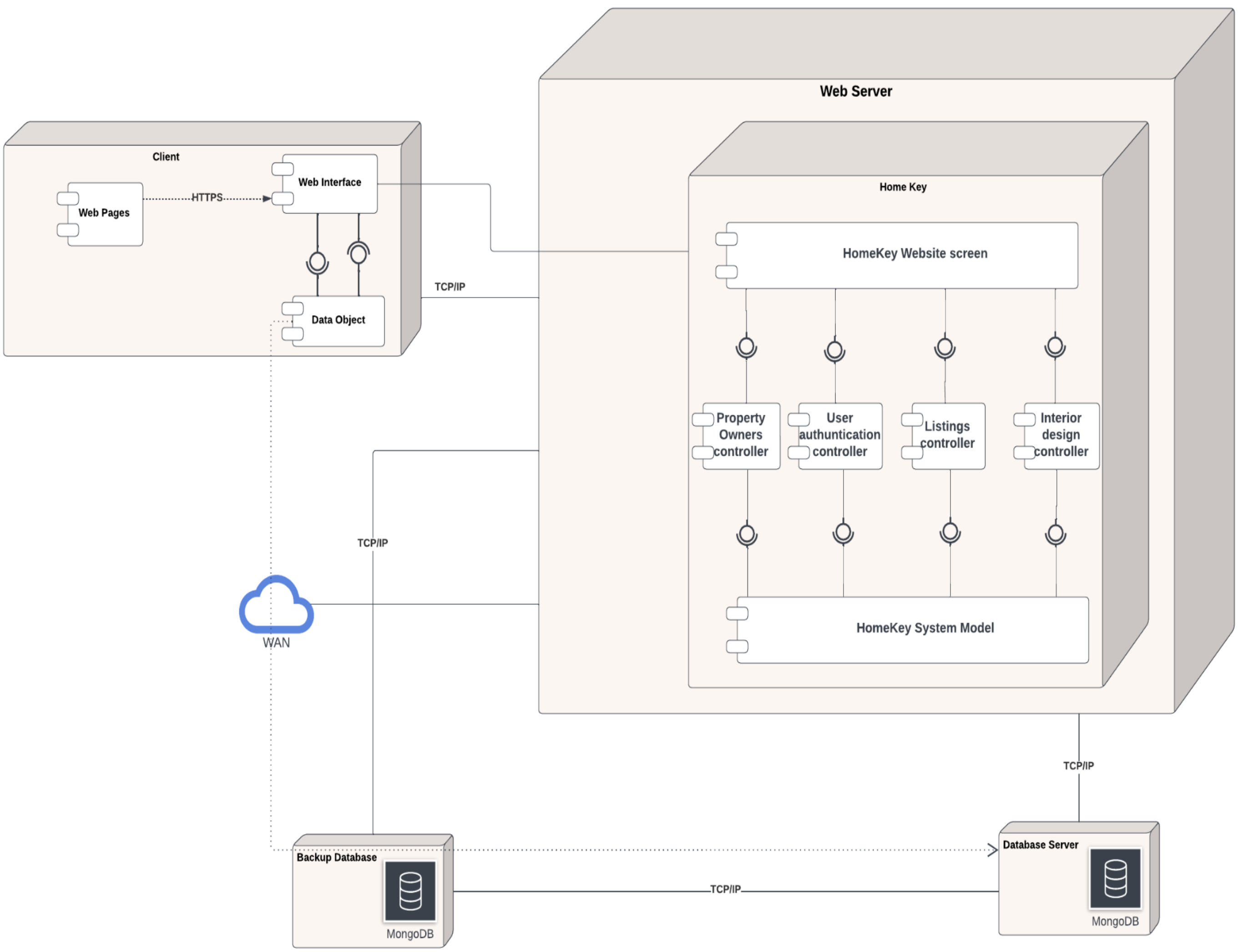
Development View - Component Diagram



Process View - Activity Diagram

1. Sign-up
2. Log-in
3. Property Listings
4. Advanced Search
5. User Profiles
6. Property Details
7. Wish List
8. Interactive Maps
9. Reviews and Ratings
10. User Support
11. Secure Messaging
12. Interior Design Services
13. Nearby Amenities
14. Inspection team

Physical View - Deployment Diagram



3. Database Schema

Entity Definition Table:

Entity	Description	Identifier	Attributes	Field Type
User	entity that contains all the details about registered users.	User_ID	- User_ID (unique)	ObjectId
			- FName	String
			- LName	Str
			- Email (unique, Required)	String
			- Password (min length : 8 digits, Required)	String
			- PhoneNo (min length : 10 digits,	Integer

			unique, Required)	
			- Role (Property Owner, Interior Designer, Seeker)	
Properties	entity that contains all details about the added properties.	Property_ID	- Property_ID (unique)	ObjectId
			- PropertyType	Array [villa, apartment, chalet, resort, studio, camping spot]
			- District	String
			- NumberOfRooms	Integer
			- NumberOfFloors	Integer
			- Price	Integer
			- MainRoad	String
			- SizeOfProperty	Integer

			- Description	String
			- Images	String
			- Location	Array
Designer Work	entity that contains all details about the added designer's work.	Work_ID	- Work_ID (unique)	ObjectId
			- YearsOfExperience	Integer
			- Description	String
			- Images	String
Review and Rating	entity that contains all the added details about the review and rating	R_ID	- R_ID (unique)	ObjectId
			- Review	String
			- Rating	Integer
Wishlist Items	entity that contains all the added wishlists from the user.	Wishlist_ID	- Wishlist_ID	ObjectId
			- Property_ID	ObjectId

			- Work_ID	ObjectId

4. Detailed Design

Since our product is developed in a classless language , we've decided not to use specific design patterns. While some websites do use them, they don't fit well with our approach using Node.js. So, we're not following any particular design patterns in our development.

5. Human Interface Design

5.1. Website Visual Identity

Primary Logo:

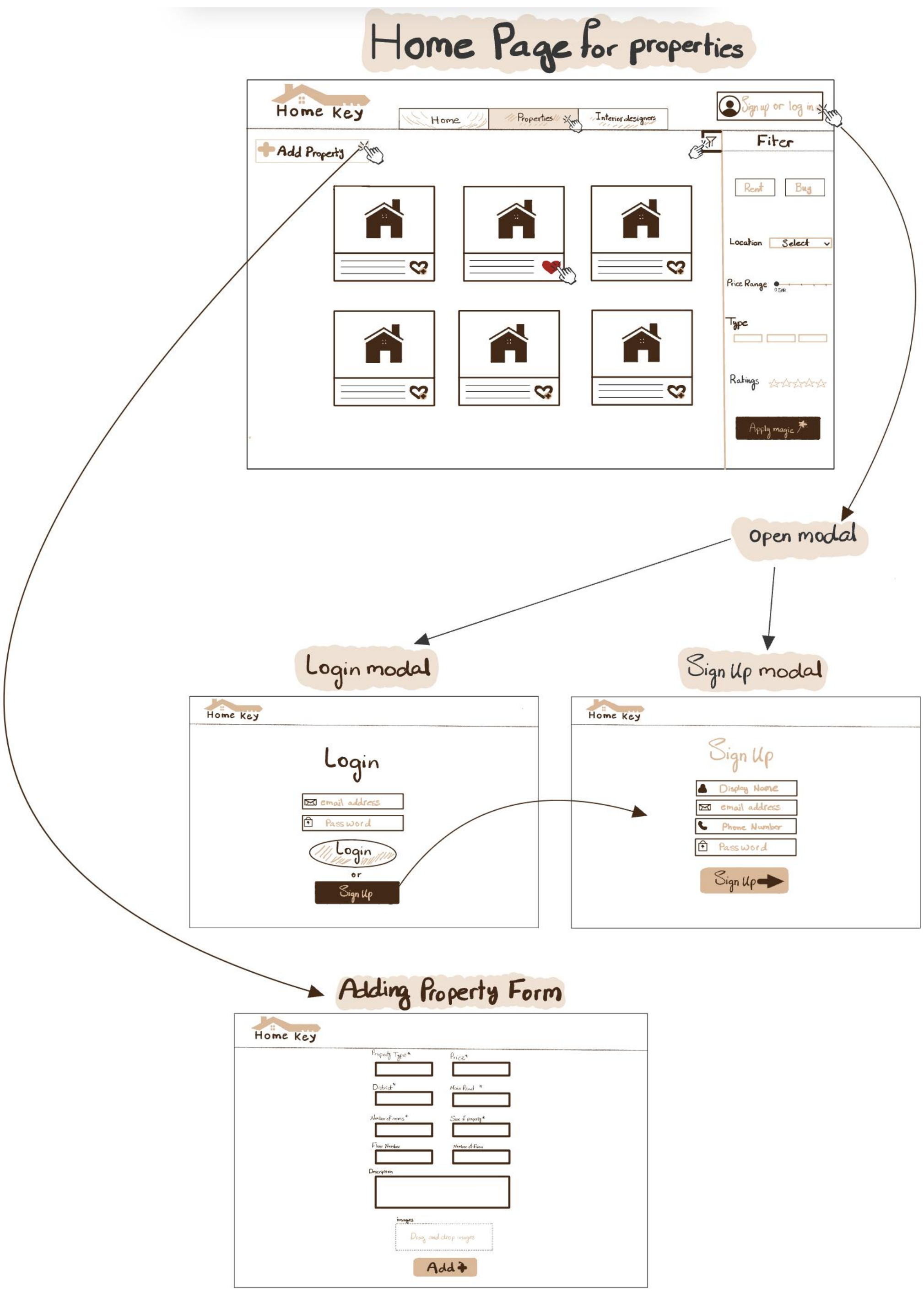
Color Palette:

Font:

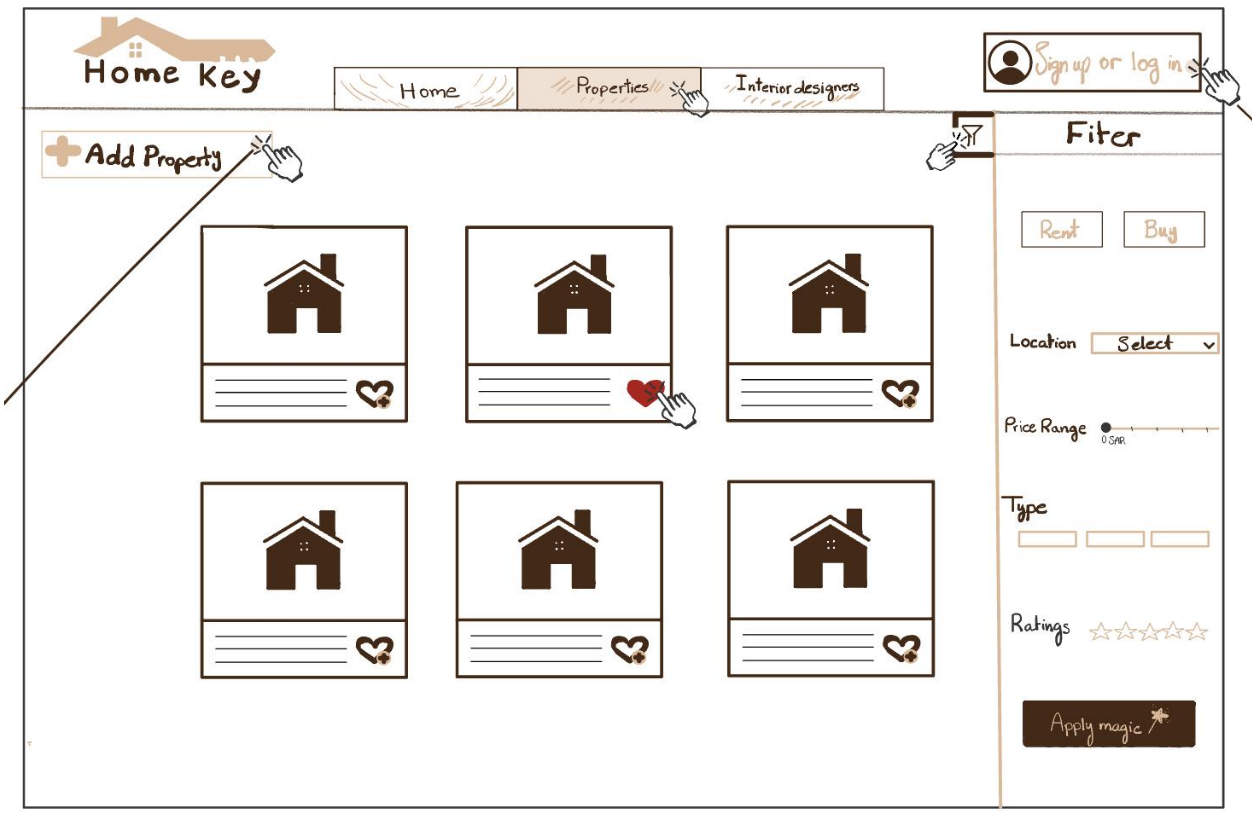
5.2 Low-fidelity Prototype

Our deliberate choice to utilize low-fidelity prototypes in the form of sketches played a pivotal role in our project's development. This hands-on approach allowed us to meticulously refine each element, guaranteeing clarity, comprehensiveness, and precision in our designs. Through these detailed sketches, we eliminated any potential ambiguity, ensuring a clear and comprehensive representation. This iterative process not only deepened our understanding of the design direction but also provided a tangible preview of what to expect in the final product. As a result, it paved the way for a polished, well-thought-out end result in the development of HomeKey.

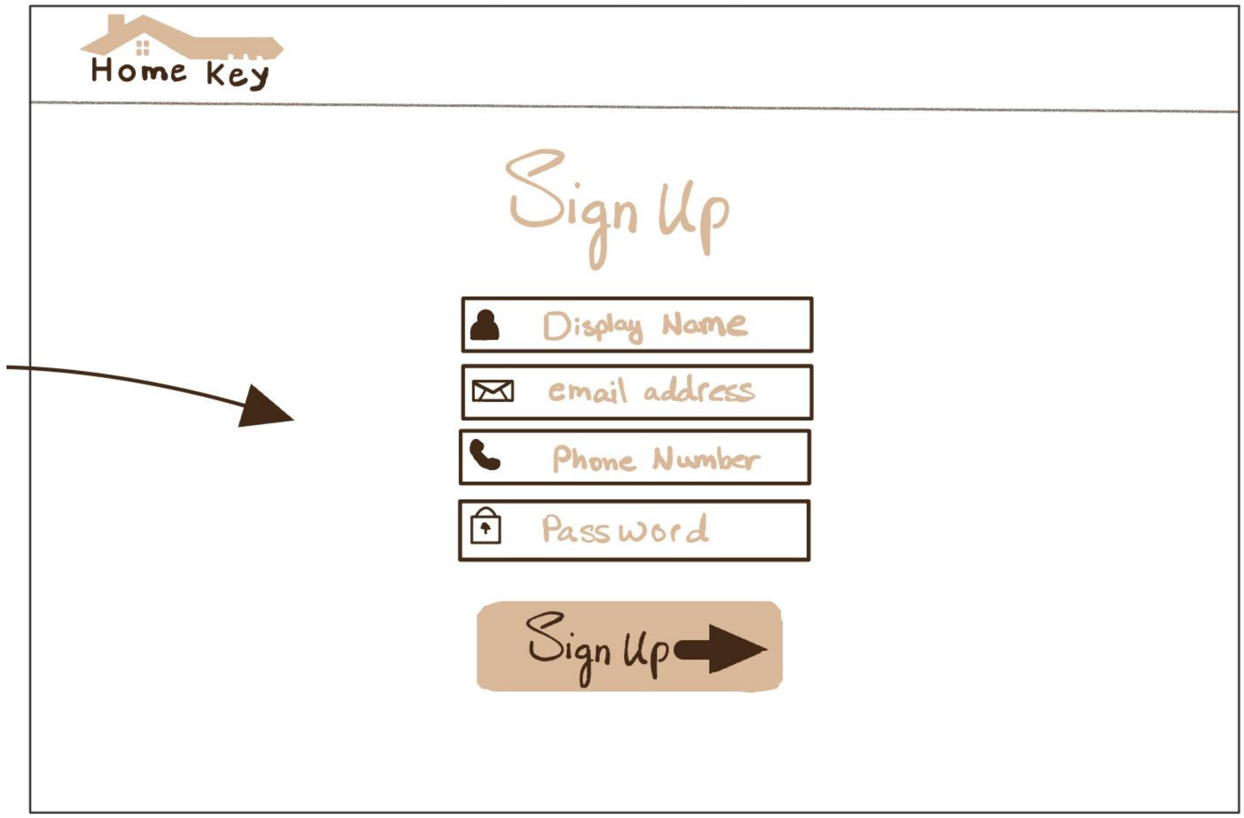
Kindly note that a close-up view of this can be found in the following pages.



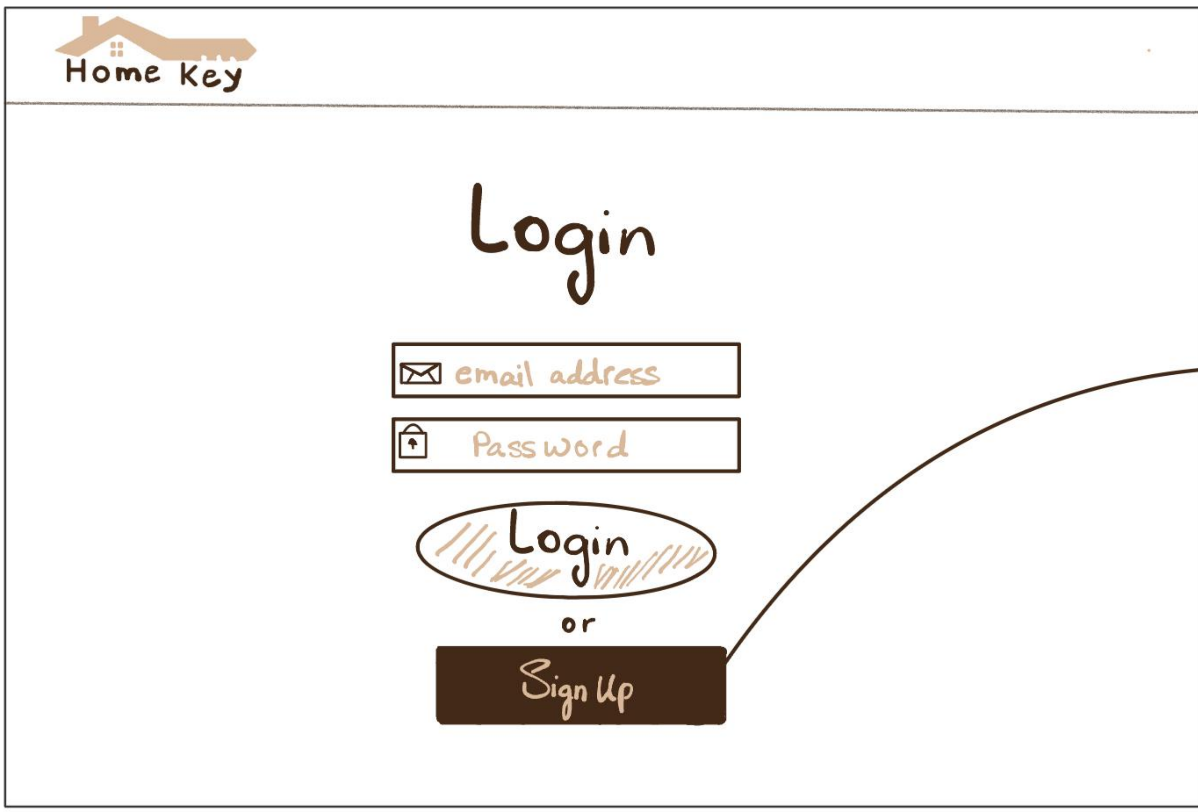
Home Page for properties



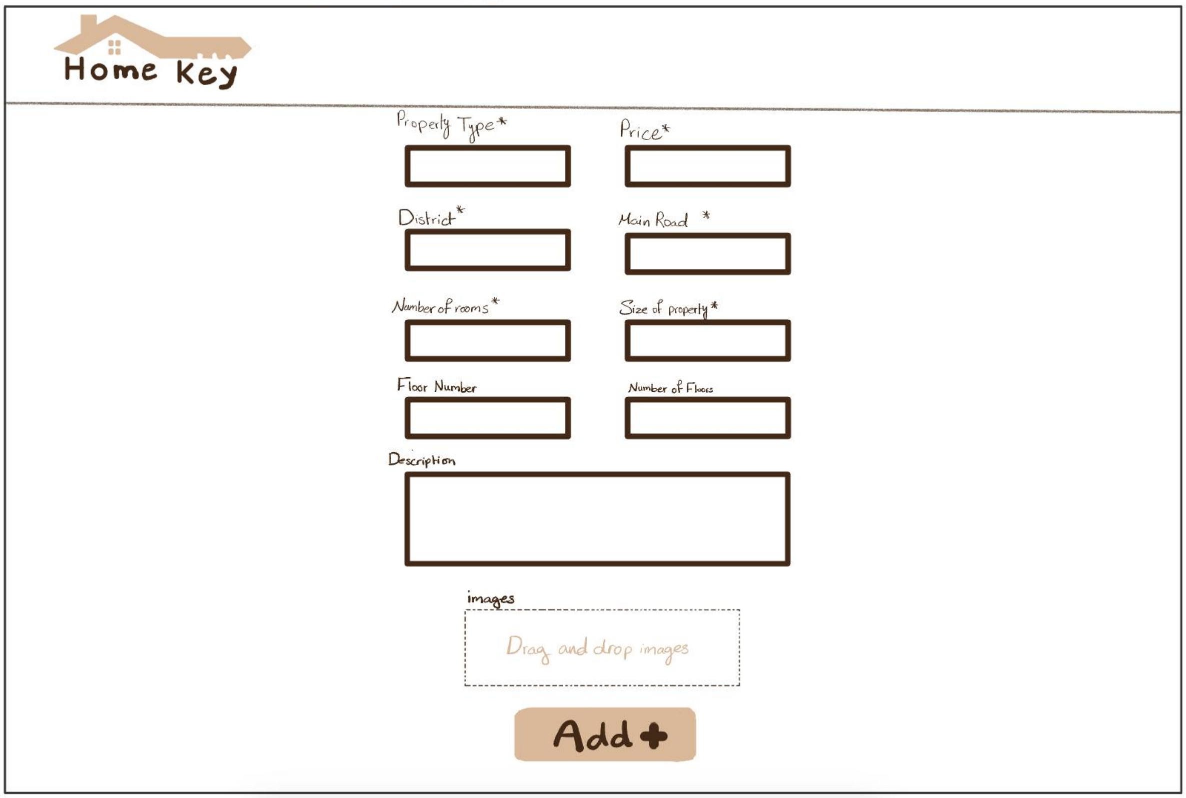
Sign Up modal



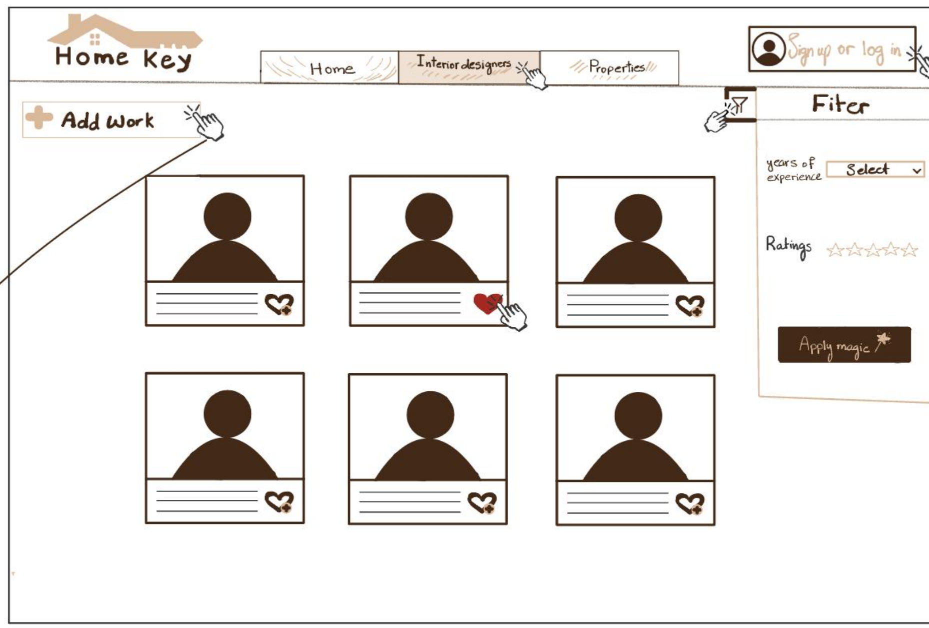
Login modal



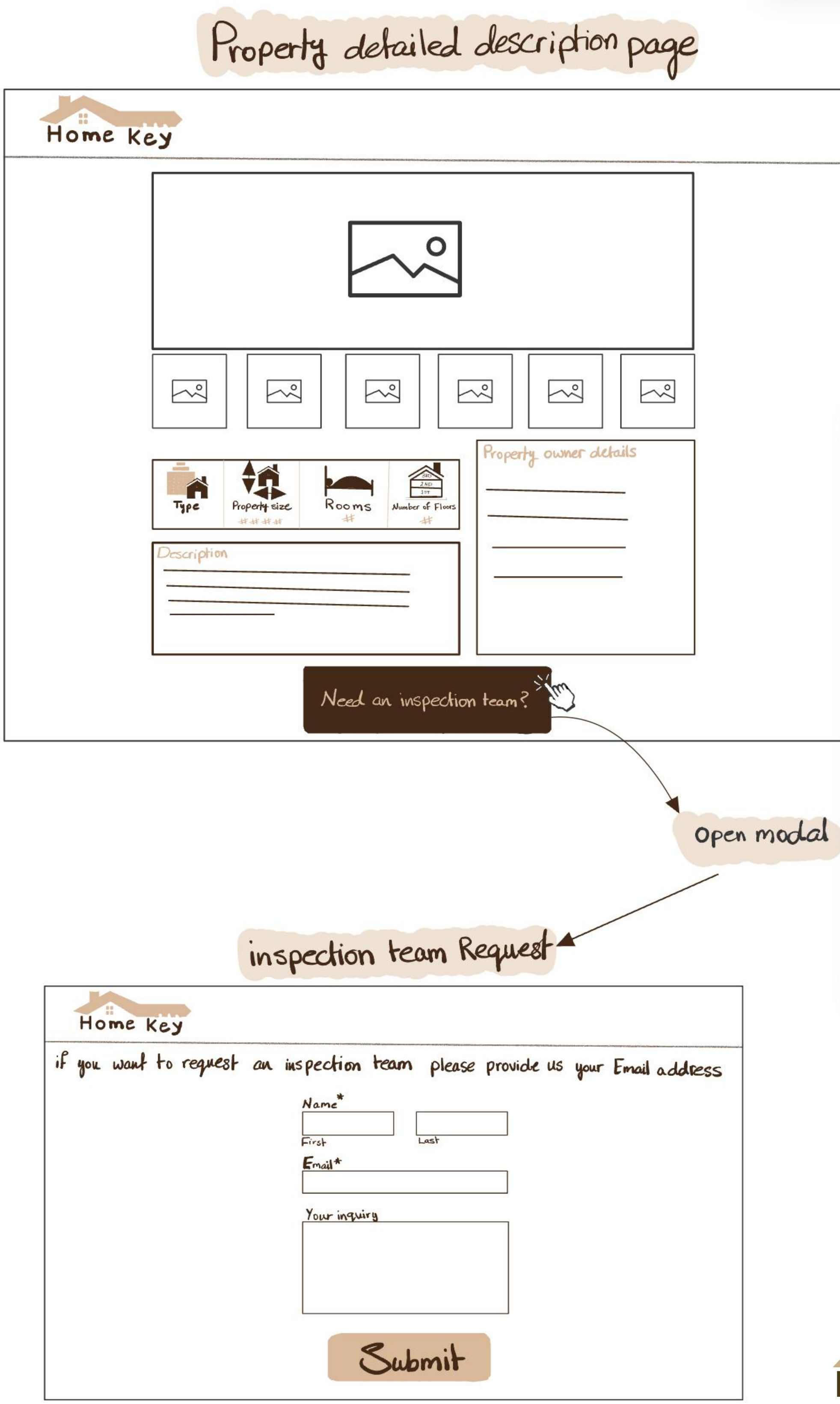
Adding Property Form



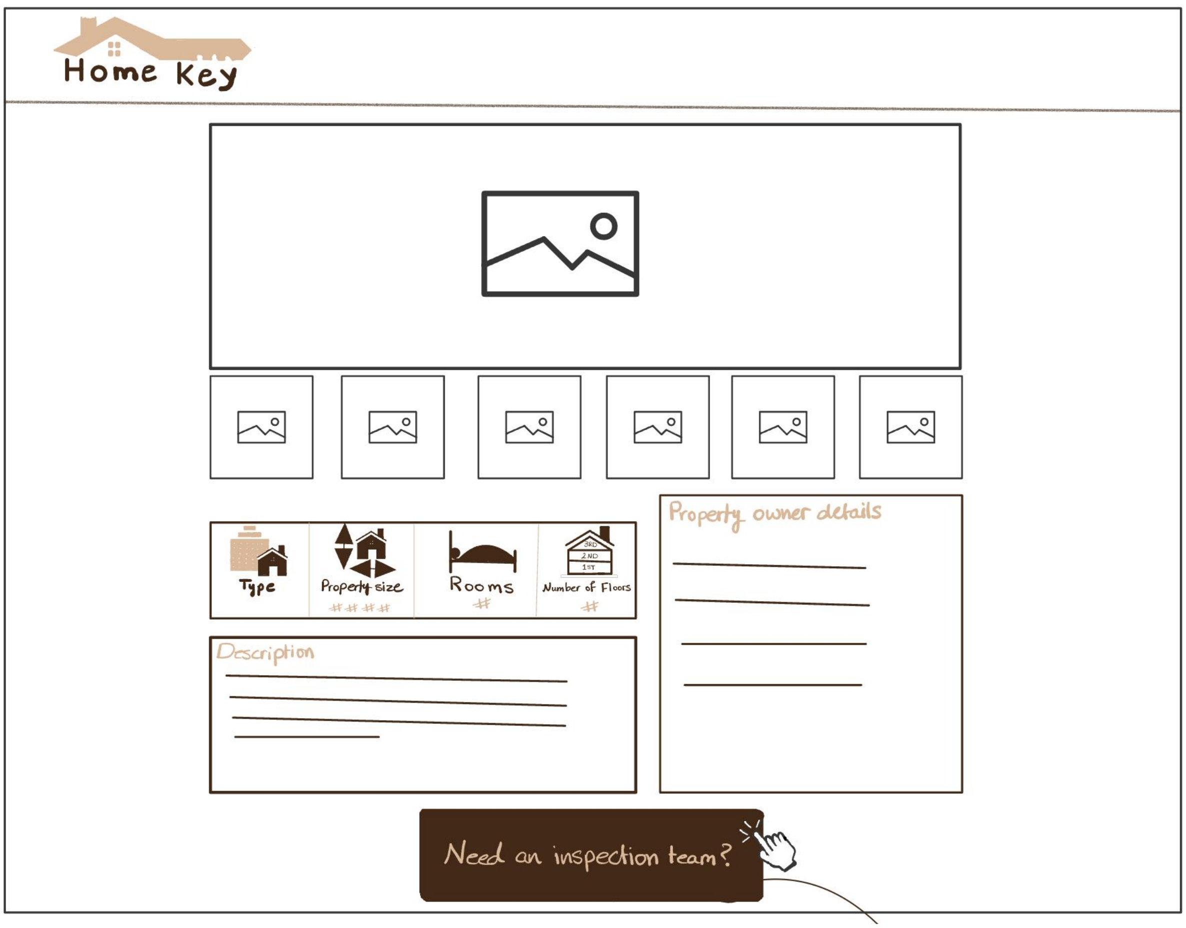
Home Page for Interior designers



Interior designers work Form


Property detailed description page
Home Key
Type
Property size
Rooms
Number of Floors
Property owner details
Description
Need an inspection team?
Open modal
inspection team Request
Home Key
if you want to request an inspection team please provide us your Email address
Name*
First
Last
Email*
Your inquiry
Submit

Property detailed description page



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inspection team Request

Home Key

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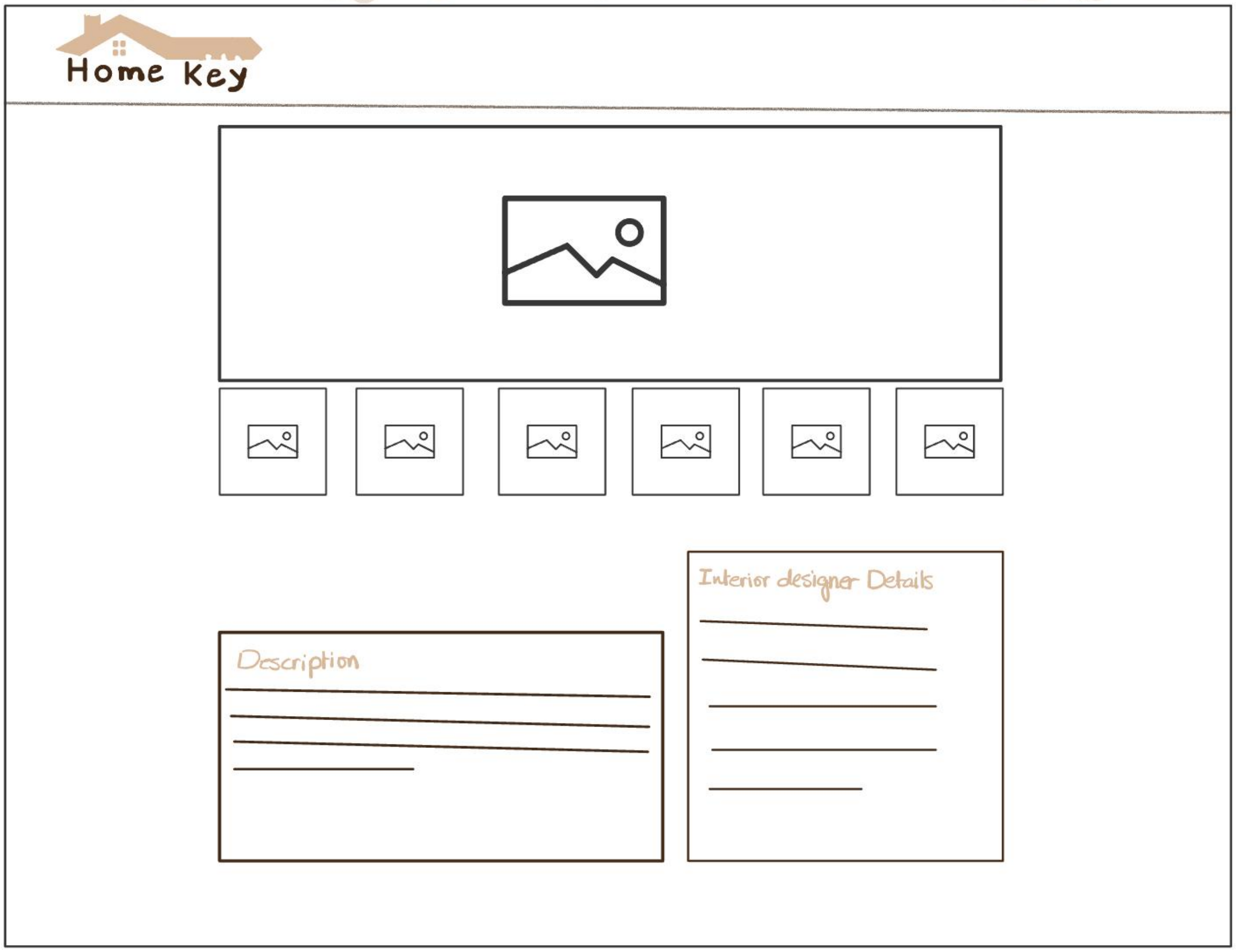
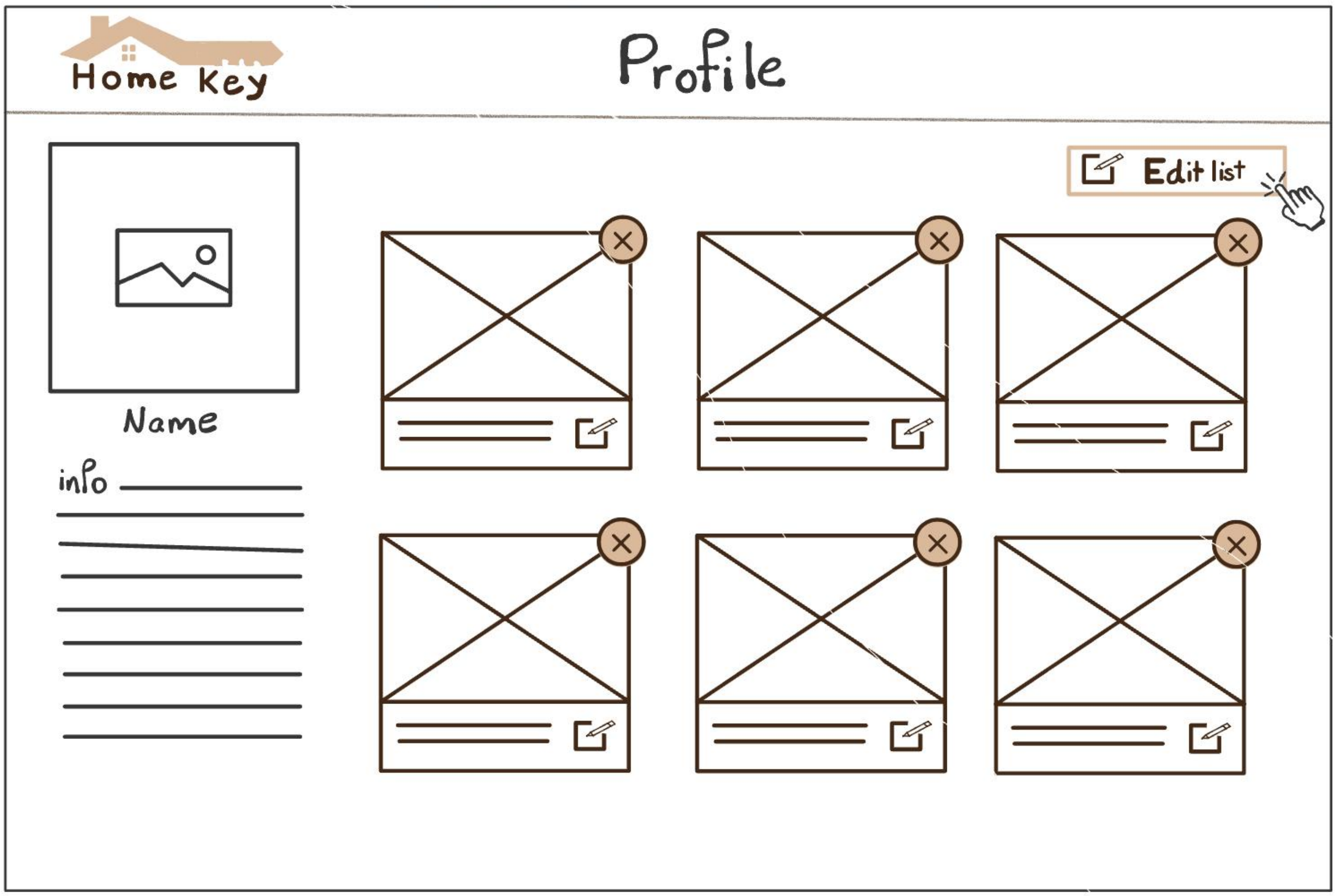
Interior designer's work detailed description page



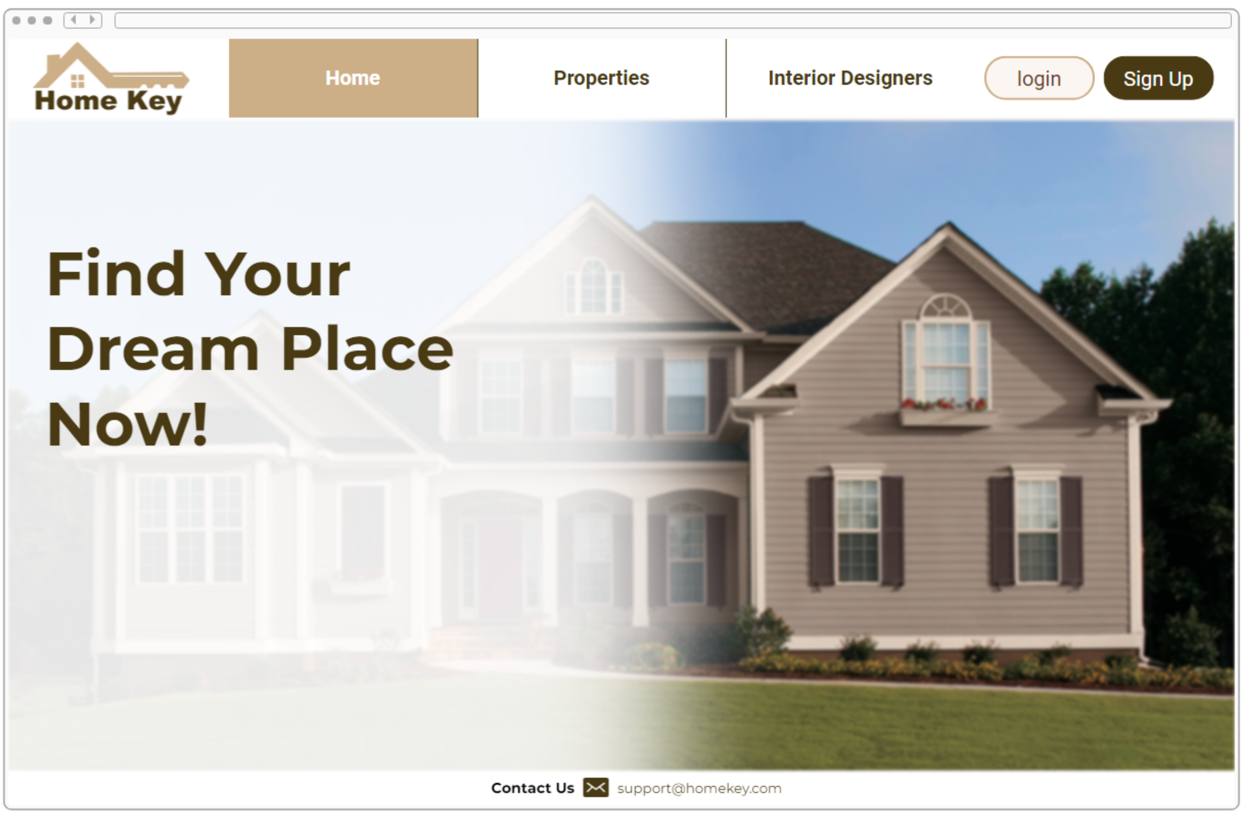
user profile and edit

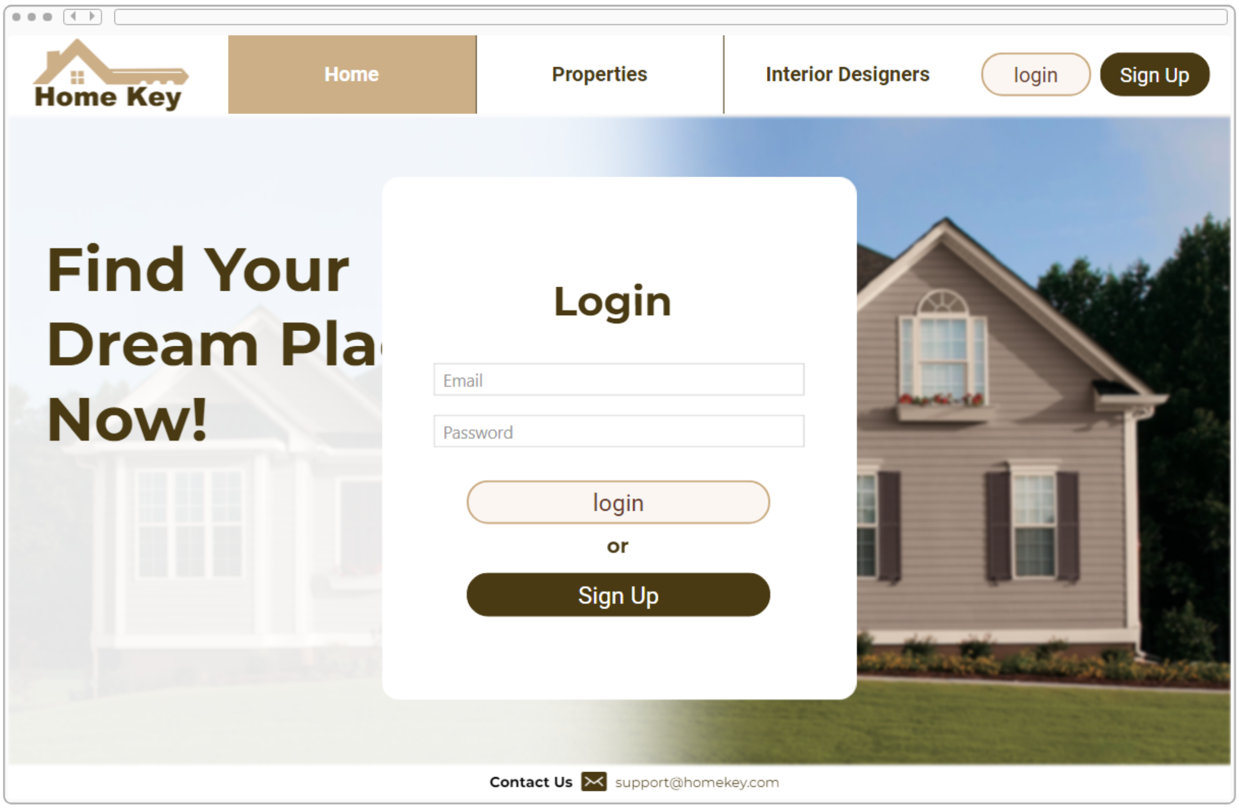


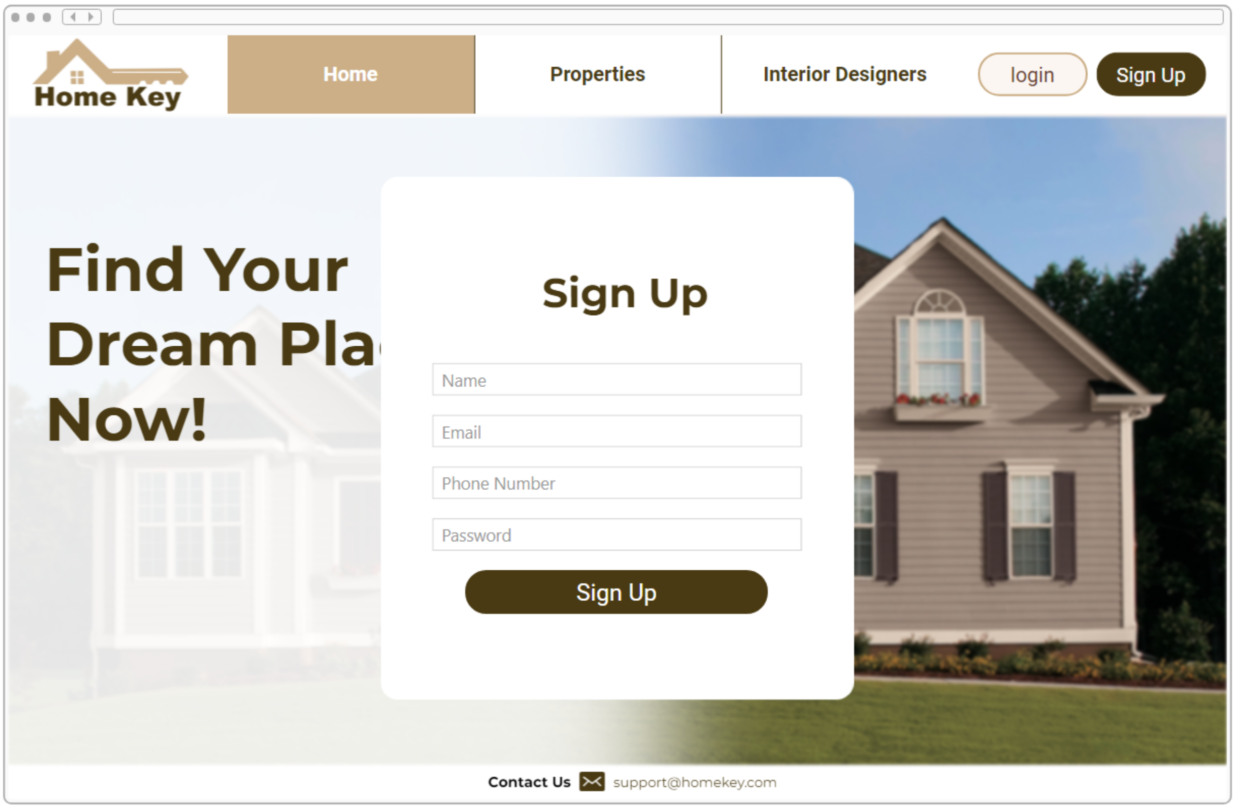
5.3 High-fidelity Prototype

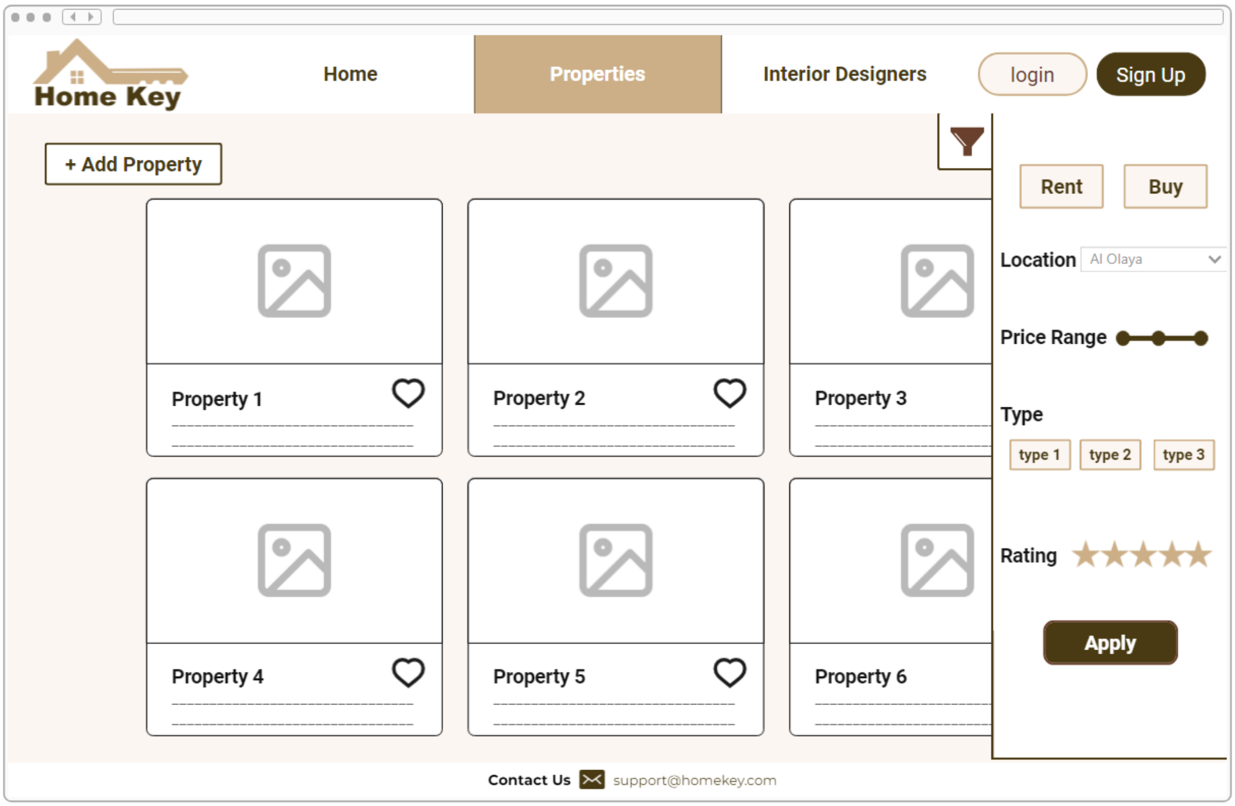
To develop our project's prototype, we employed Proto.io, a powerful prototyping tool that facilitated the creation of a dynamic and interactive model. This platform not only enabled us to design the user interface but also allowed us to simulate user interactions, providing a comprehensive preview of the final product's functionality. With Proto.io, we were able to bring our vision to life, setting a new standard for the real estate industry.

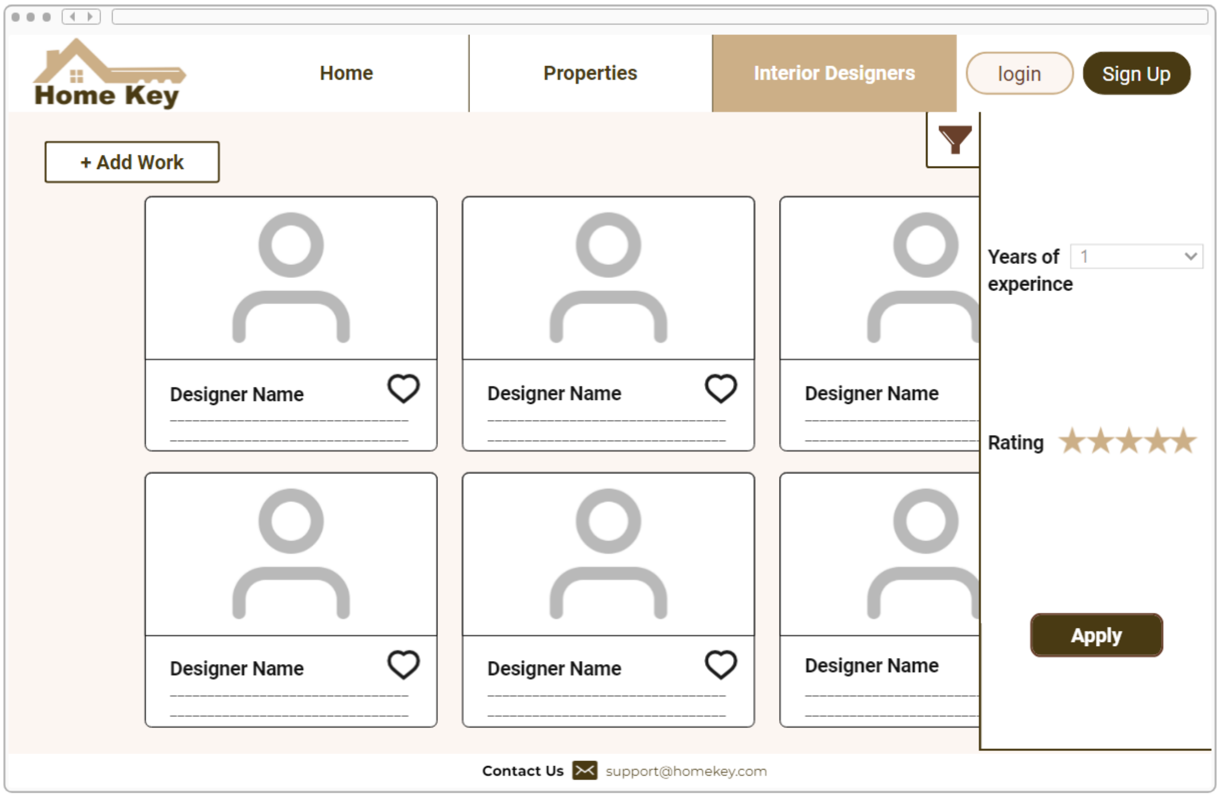
HomeKey Prototype Link: https://pr.to/SU8R9K/

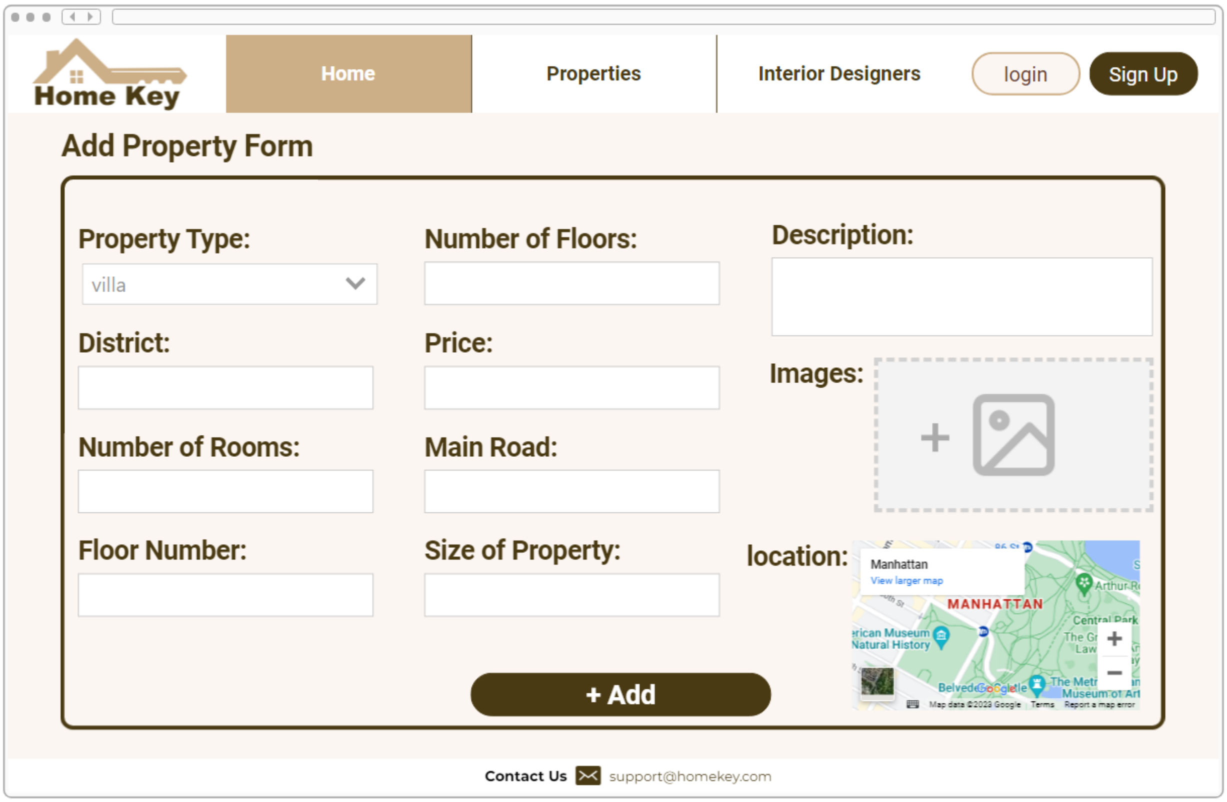


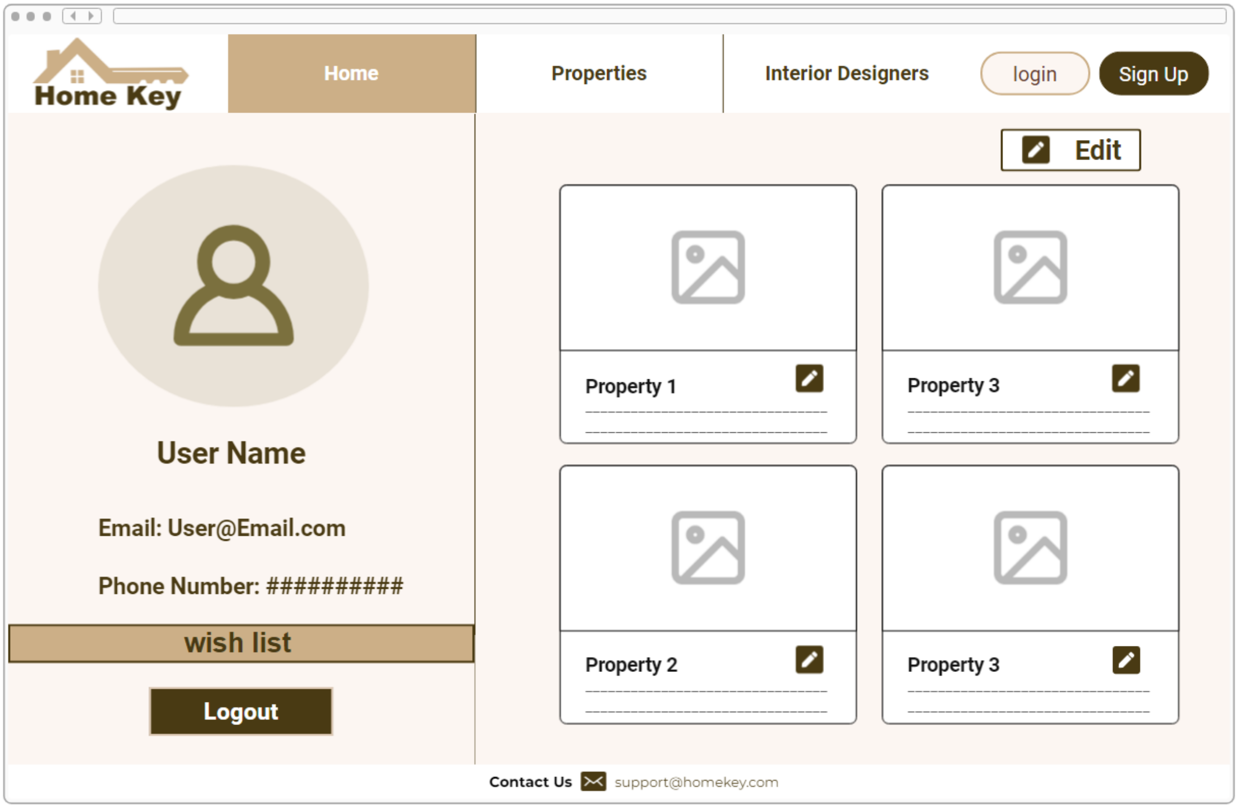

Home Key
Home
Properties
Interior Designers
login
Sign Up
Find Your
Dream Pla
Now!
Login
Email
Password
login
or
Sign Up
Contact Us
support@homekey.com


Home Key
Home
Properties
Interior Designers
login
Sign Up
Find Your Dream Pla
Now!
Sign Up
Name
Email
Phone Number
Password
Sign Up
Contact Us
support@homekey.com

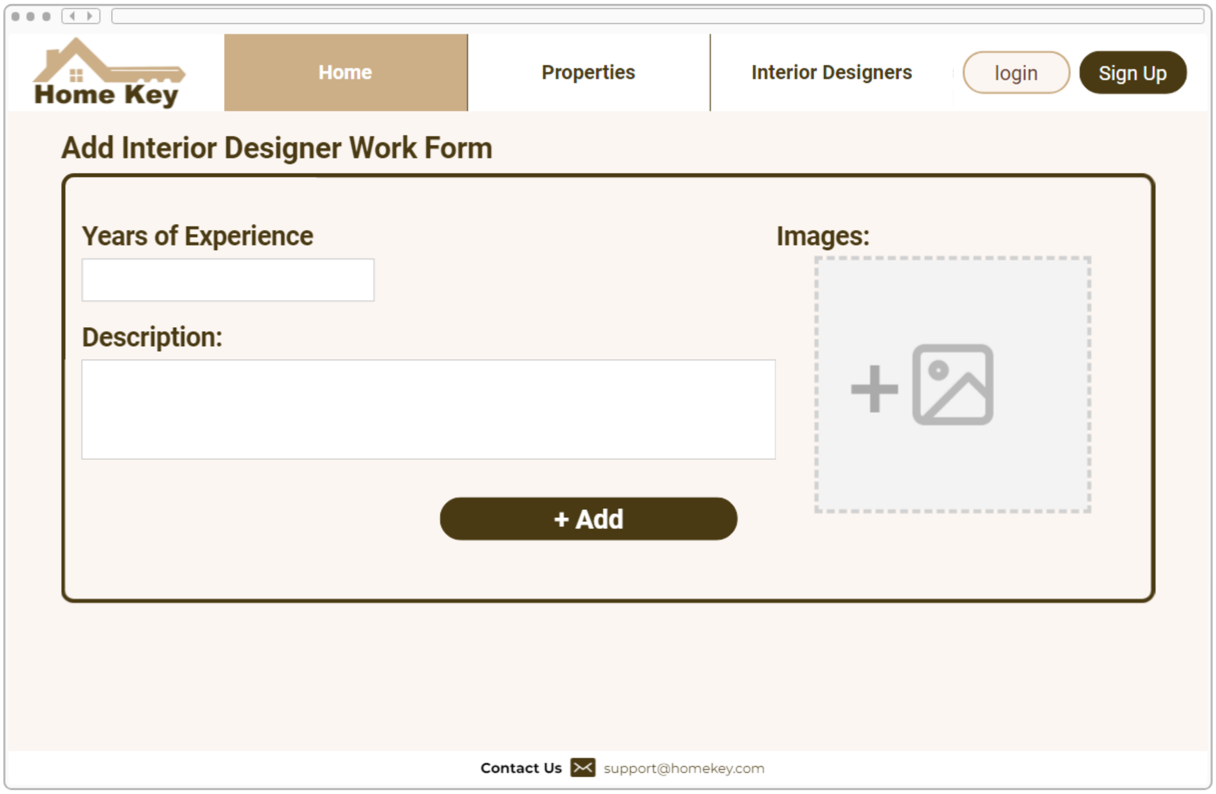

Home Key
Home
Properties
Interior Designers
login
Sign Up
+ Add Property
Property 1
Property 2
Property 3
Property 4
Property 5
Property 6
Rent
Buy
Location
Al Olaya
Price Range
Type
type 1
type 2
type 3
Rating
Apply
Contact Us
support@homekey.com

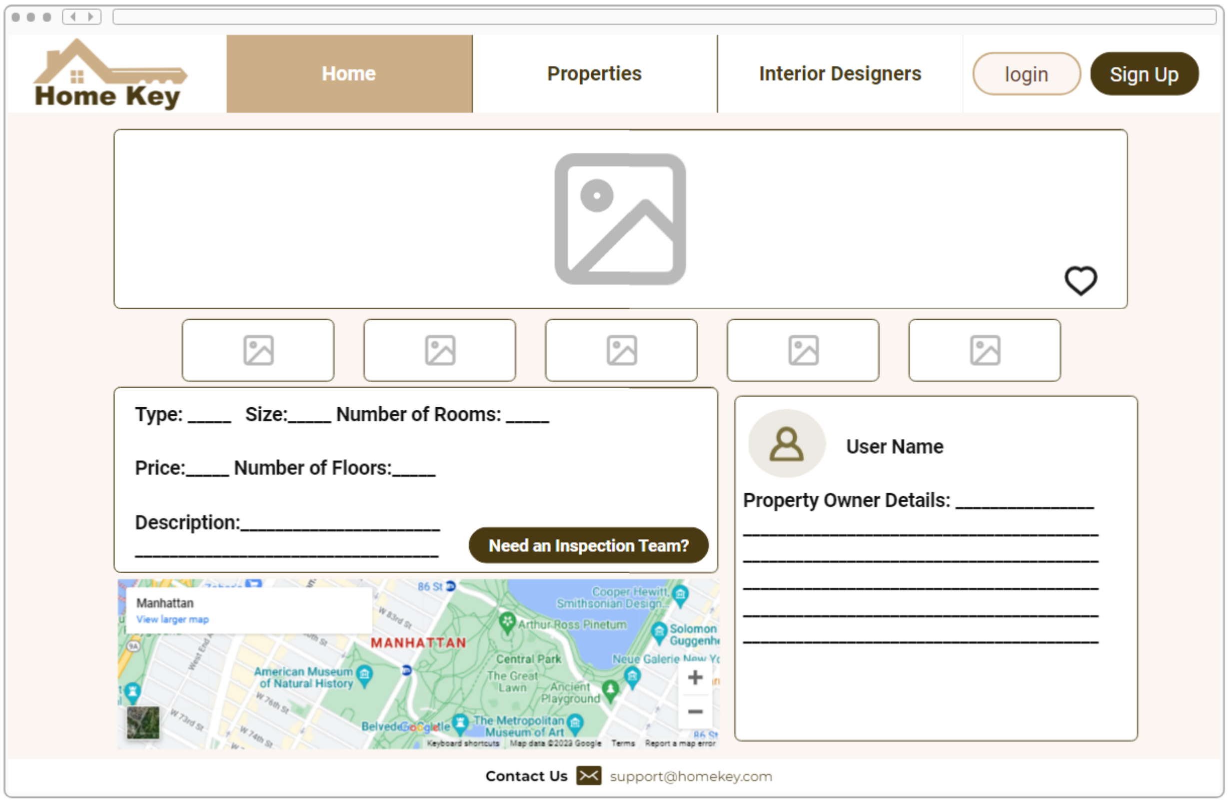

Home Key
Home
Properties
Interior Designers
login
Sign Up
+ Add Work
Designer Name
Designer Name
Designer Name
Designer Name
Designer Name
Designer Name
Years of experince
1
Rating
Apply
Contact Us
support@homekey.com


Home Key
Home
Properties
Interior Designers
login
Sign Up
Add Property Form
Property Type:
villa
Number of Floors:
Description:
District:
Price:
Images:
Number of Rooms:
Main Road:
Floor Number:
Size of Property:
location:
Manhattan
View larger map
MANHATTAN
Central Park
+ Add
Contact Us
support@homekey.com



Home Key
Home
Properties
Interior Designers
login
Sign Up
Edit
User Name
Email: User@Email.com
Phone Number: ##########
wish list
Logout
Property 1
Property 3
Property 2
Property 3
Contact Us
support@homekey.com


Home Key
Home
Properties
Interior Designers
login
Sign Up
Add Interior Designer Work Form
Years of Experience
Images:
Description:
+ Add
Contact Us
support@homekey.com



Home Key
Home
Properties
Interior Designers
login
Sign Up
Type: _____ Size:_____ Number of Rooms: _____
Price:_____ Number of Floors:_____
Description:____________________
____
Need an Inspection Team?
User Name
Property Owner Details: ______________
Manhattan
View larger map
MANHATTAN
Central Park
American Museum of Natural History
The Metropolitan Museum of Art
Contact Us
support@homekey.com

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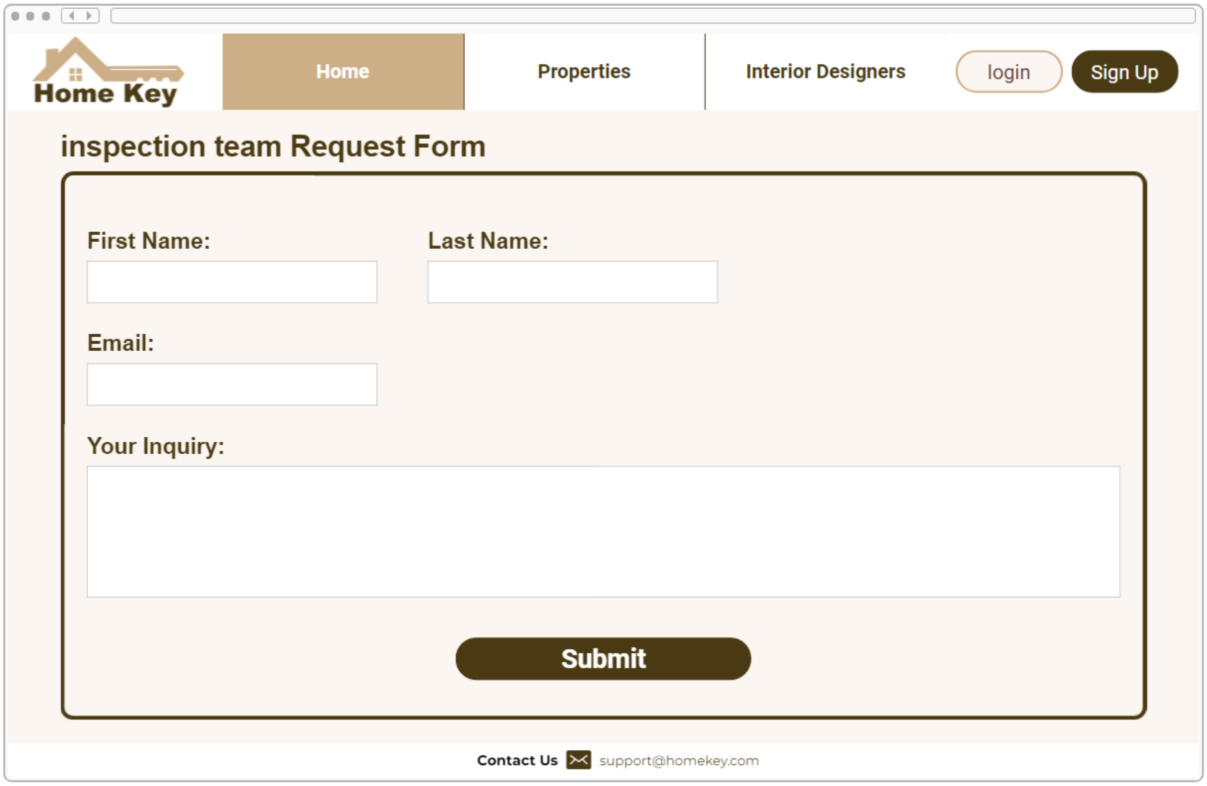
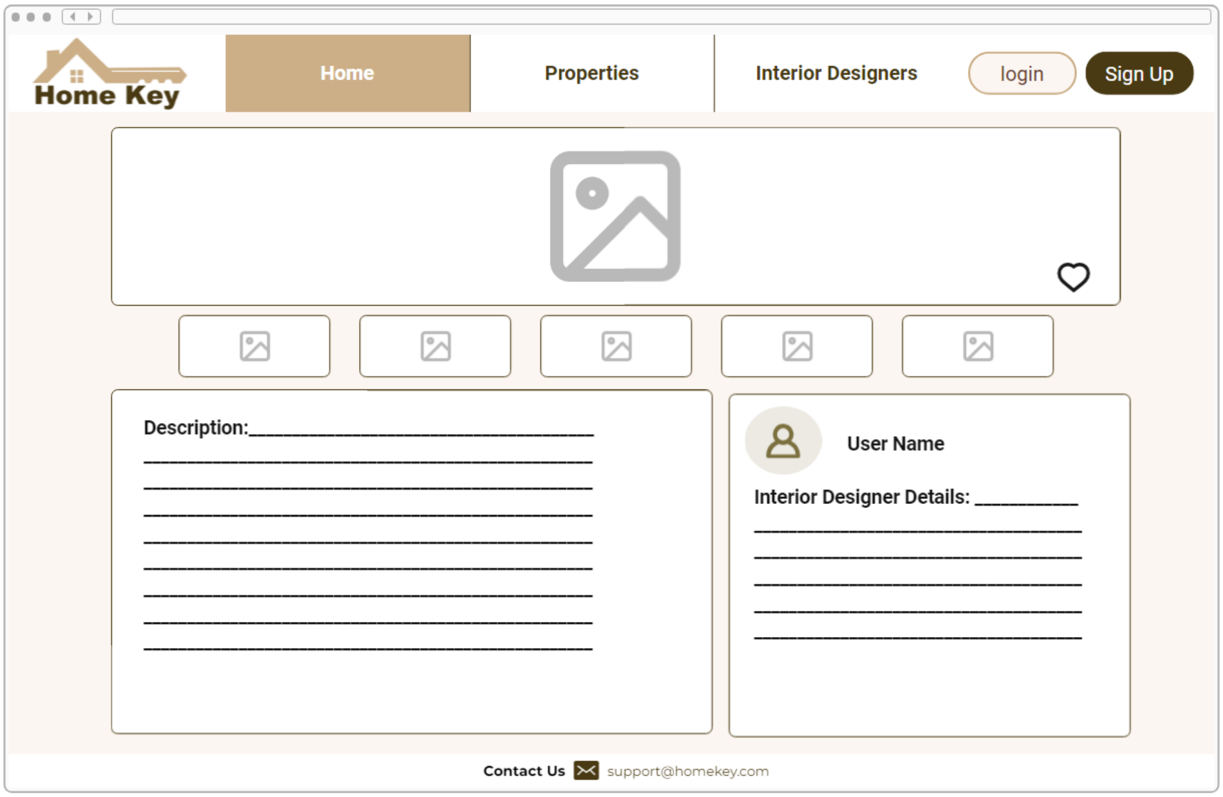


Home Key


Home Key
Home
Properties
Interior Designers
login
Sign Up
Description:
User Name
Interior Designer Details:
Contact Us
support@homekey.com

6. Heuristic Evaluation

The high-fidelity prototypes were subjected to a heuristic evaluation in which the participant was instructed to navigate through the screens and complete a form based on her personal opinion and assessment of screen usability.

Evaluator: Joudy Koujan
Browser: Windows, Chrome.

Rule of Thumb	Is this rule being applied? How so?	Is this rule being violated? How so?	How can this rule further improve, usability, utility, and desirability?
1. visibility of system status	Yes, clear status feedback.	No violations.	-
2. Match between the system and the real world	Yes, icons match real-world concepts.	No violations.	-
3. User control and freedom	Yes, users have control.	No undo feature.	Add an undo button.
4. Consistency and Standards	Yes, consistent design elements.	No violations.	Maintainability
5. Error prevention	Yes, errors are avoided	No violations.	-
6. Recognition rather than recall	Yes, icons were easy to recognize	No violations.	-
7. Flexibility and Efficiency of Use	Yes, it is flexible and efficient, they provided filtration.	No violations.	-
8. Aesthetic and Minimalist	Yes, clean and appealing design.	No violations.	-
9. Help users	Needs improved	Lack of error	Clear errors with

recognize, diagnose, and recover from errors	error communication.	feedback.	solutions.
10. Help and documentation	yes, there was an email provided to contact for support.	No violations.	-

After looking at the answers, we analyzed that our website is doing well in most areas, such as showing clear status updates, using familiar icons, and avoiding errors. Our design is clean and user-friendly, and we offer some help via email. However, we should add an undo option for users to have more control, and it's important to improve error messages with clear solutions. These changes will make our website even more user-friendly and helpful.

Home Key

SDS

SOFTWARE DESIGN SPECIFICATION PROCESS

Iterations:

First Iteration:

In the first iteration, our primary objective was to identify an optimal architectural style, and the creation of both low and high-fidelity prototypes. After careful thinking, we chose the three-tier architecture as the foundation for HomeKey, a decision supported by a multitude of compelling reasons outlined above. For the low-fidelity prototype, we employed detailed sketches to capture the essence of the design, while the high-fidelity counterpart was meticulously crafted using Proto.io, ensuring a seamless and user-friendly experience. Additionally, we developed a comprehensive data model diagram to effectively represent the structure and relationships within our database, further enhancing the system's functionality and performance.

Second Iteration:

In the second stage of our project, we applied the 4+1 views model, a flexible framework that allows us to analyze HomeKey from various perspectives. This approach grants us a comprehensive understanding of both its structure and operation. Within this model, we explore four distinct views: the logical view, where we meticulously crafted the Data Model Diagram, the process view, represented by an Activity Diagram, the development view, elucidated through a Component Diagram, and finally, the physical view, captured in the Deployment Diagram. These views and their corresponding diagrams collectively contribute to a well-rounded comprehension of HomeKey's architecture and operational flow.

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Home Key